



CHAPTER

1

FUNCTIONS & LIMITS

Function:

A function f from a set X to a set Y is a rule or a correspondence that assigns to each element x in X a unique element y in Y . The set X is called domain of f . The set of corresponding elements y in Y is called the range of f .

Types of Functions:

Some important types of functions are given below.

Algebraic Functions:

Algebraic functions are those functions, which are defined by algebraic expressions we classify algebraic functions as follows.

(i) Polynomial Function:

A function P of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0$$

for all x , where the coefficient $a_n, a_{n-1}, a_{n-2}, \dots, a_2, a_1, a_0$ are real numbers and the exponents are non-negative integers, is called a polynomial function.

The domain and range of $P(x)$ are, in general, subsets of real numbers.

If $a_n \neq 0$, then $P(x)$ is called a polynomial function of degree n and a_n is the leading coefficient of $P(x)$.

(ii) Linear Function:

If the degree of a polynomial function is 1, then it is called a linear function. A linear function is of the form $f(x) = ax + b$ ($a \neq 0$), a, b are real numbers.

Its domain and range are the set of real numbers.

(iii) Identity Function:

For any set X , a function $I: X \rightarrow X$ of the form $I(x) = x \forall x \in X$, is called an identity function. Its domain and range is the set X itself.

(iv) Constant Function:

Let X and Y be sets of real numbers. A function $C: X \rightarrow Y$ defined by

$C(x) = a, \forall x \in X, a \in Y$ and fixed is called a constant function.



(v) Rational Function:

A function $R(x)$ of the form $\frac{P(x)}{Q(x)}$, where both $P(x)$ and $Q(x)$ are polynomial functions and $Q(x) \neq 0$, is called Rational Function. The domain of a rational function $R(x)$ is the set of all real numbers x for which $Q(x) \neq 0$.

Exponential Function:

A function in which the variable appears as exponent (power) is called an exponential function. The functions $y = e^{ax}$, $y = e^x$, $y = 2^x$ etc are called exponential functions of x .

Logarithmic Function:

If $x = a^y$, then $y = \log_a x$, where $a > 0$, $a \neq 1$ is called Logarithmic Function of x .

- (i) If $a = 10$, then we have $\log_{10} x$ which is known as the common logarithm of x .
- (ii) If $a = e$, then we have $\log_e x$ which is known as the natural logarithm of x .

Explicit Function:

If y is easily expressed in terms of the independent variable x , then y is called an explicit function of x .

For example (i) $y = x^2 + 2x - 1$ (ii) $y = \sqrt{x - 1}$ are explicit functions of x .
Symbolically it can be written as $y = f(x)$.

Implicit Function:

If x and y are so mixed up and y cannot be expressed in terms of the independent variable x , then y is called an implicit function of x .

For example

$$(i) \quad x^2 + xy + y^2 = 2 \quad (ii) \quad \frac{xy^2 - y + 9}{xy} = 1$$

are implicit functions of x and y . Symbolically it is written as $f(x, y) = 0$

Parametric Functions:

Some times a curve is described by expressing both x and y as functions of a third variable ' t ' or ' θ ' which is called a parameter. The equations of the type $x = f(t)$ and $y = g(t)$ are called the parametric equations of the curve.

The functions of the form

$$(i) \quad x = at^2, y = at \quad (ii) \quad x = a \cos t, y = a \sin t$$

$$(iii) \quad x = a \cos \theta, y = b \sin \theta \quad (iv) \quad x = a \sec \theta, y = b \tan \theta$$

are called parametric functions. Here the variable ' t ' or ' θ ' is a parameter.

Hyperbolic Functions:

- (i) $\sinh x = \frac{e^x - e^{-x}}{2}$ is called hyperbolic sine function. Its domain and range are the set of real numbers.
- (ii) $\cosh x = \frac{e^x + e^{-x}}{2}$ is called hyperbolic cosine function. Its domain is the set of all real numbers and the range is the set of all numbers in the interval $[1, +\infty)$.
- (iii) $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- (iv) $\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$
- (v) $\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$
- (vi) $\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$

Inverse Hyperbolic Functions:

- (i) $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$, for all x
- (ii) $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$, $x \geq 1$
- (iii) $\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$, $|x| < 1$
- (iv) $\coth^{-1} x = \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right)$, $|x| < 1$
- (v) $\operatorname{sech}^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1-x^2}}{x}\right)$, $0 < x \leq 1$
- (vi) $\operatorname{cosech}^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right)$, $x \neq 0$

Even Function:

A function $f(x)$ is said to be an even if $f(-x) = f(x)$, for every number x in the domain of f .

Odd Function:

A function $f(x)$ is said to be an odd if $f(-x) = -f(x)$, for every number x in the domain of f .



EXERCISE 1.1

Q.1 Given that:

(a) $f(x) = x^2 - x$

(b) $f(x) = \sqrt{x+4}$

Find: (i) $f(-2)$

(ii) $f(0)$

(iii) $f(x-1)$

(iv) $f(x^2+4)$

Solution:

(a) $f(x) = x^2 - x$

(i) $f(-2) = (-2)^2 - (-2)$
 $= 4 + 2 = 6$ Ans.

(ii) $f(0) = (0)^2 - 0$
 $= 0$ Ans.

(iii) $f(x-1) = (x-1)^2 - (x-1)$
 $= x^2 - 2x + 1 - x + 1$
 $= x^2 - 3x + 2$ Ans.

(iv) $f(x^2+4) = (x^2+4)^2 - (x^2+4)$
 $= x^4 + 8x^2 + 16 - x^2 - 4$
 $= x^4 + 7x^2 + 12$ Ans.

(b) $f(x) = \sqrt{x+4}$

(i) $f(-2) = \sqrt{-2+4} = \sqrt{2}$ Ans.

(ii) $f(0) = \sqrt{0+4} = \sqrt{4} = 2$ Ans.

(iii) $f(x-1) = \sqrt{x-1+4} =$

(iv) $f(x^2+4) = \sqrt{x^2+4+4} = \sqrt{x^2+8}$ Ans.

Q.2 Find $\frac{f(a+h)-f(a)}{h}$ and simplify where,

(i) $f(x) = 6x - 9$

(ii) $f(x) = \sin x$

(iii) $f(x) = x^3 + 2x^2 - 1$

(iv) $f(x) = \cos x$

Solution:

(i) $f(x) = 6x - 9$

$f(a+h) = 6(a+h) - 9$

$f(a) = 6a - 9$

$f(a) = 6a - 9$



$$\begin{aligned}
 \frac{f(a+h) - f(a)}{h} &= \frac{6a + 6h - 9 - (6a - 9)}{h} \\
 &= \frac{6a + 6h - 9 - 6a + 9}{h} \\
 &= \frac{6h}{h} \\
 &= 6 \text{ Ans.}
 \end{aligned}$$

(ii) $f(x) = \sin x$ (Lahore Board 2008)

$$f(a+h) = \sin(a+h)$$

$$f(a) = \sin a$$

$$\begin{aligned}
 \frac{f(a+h) - f(a)}{h} &= \frac{\sin(a+h) - \sin a}{h} \\
 &= \frac{2 \cos\left(\frac{a+h+a}{2}\right) \sin\left(\frac{a+h-a}{2}\right)}{h} \quad \because \sin p - \sin q = 2 \cos\left(\frac{p+q}{2}\right) \sin\left(\frac{p-q}{2}\right) \\
 &= \frac{2}{h} \cos\left(\frac{2a+h}{2}\right) \sin\left(\frac{h}{2}\right) \\
 &= \frac{2}{h} \cos\left(\frac{2a}{2} + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right) \\
 &= \frac{2}{h} \cos\left(a + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right) \text{ Ans.}
 \end{aligned}$$

(iii) $f(x) = x^3 + 2x^2 - 1$

$$f(a+h) = (a+h)^3 + 2(a+h)^2 - 1$$

$$= a^3 + h^3 + 3a^2h + 3ah^2 + 2(a^2 + 2ah + h^2) - 1 \quad \because [(a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2]$$

$$= a^3 + h^3 + 3a^2h + 3ah^2 + 2a^2 + 4ah + 2h^2 - 1$$

$$f(a) = a^3 + 2a^2 - 1$$

$$\begin{aligned}
 \frac{f(a+h) - f(a)}{h} &= \frac{a^3 + h^3 + 3a^2h + 3ah^2 + 2a^2 + 4ah + 2h^2 - 1 - (a^3 + 2a^2 - 1)}{h} \\
 &= \frac{a^3 + h^3 + 3a^2h + 3ah^2 + 2a^2 + 4ah + 2h^2 - 1 - a^3 - 2a^2 + 1}{h} \\
 &= \frac{h^3 + 3a^2h + 3ah^2 + 4ah + 2h^2}{h} \\
 &= \frac{h(h^2 + 3a^2 + 3ah + 4a + 2h)}{h} \\
 &= h^2 + (3a+2)h + 3a^2 + 4a \text{ Ans.}
 \end{aligned}$$



$$(iv) \quad f(x) = \cos x$$

$$f(a+h) = \cos(a+h)$$

$$f(a) = \cos a$$

$$\frac{f(a+h) - f(a)}{h} = \frac{\cos(a+h) - \cos a}{h}$$

$$= \frac{-2 \sin\left(\frac{a+h+a}{2}\right) \sin\left(\frac{a+h-a}{2}\right)}{h}$$

$$\because \left[\cos p - \cos q = -2 \sin\left(\frac{p+q}{2}\right) \sin\left(\frac{p-q}{2}\right) \right]$$

$$= \frac{-2}{h} \sin\left(\frac{2a+h}{2}\right) \sin\left(\frac{h}{2}\right)$$

$$= \frac{-2}{h} \sin\left(\frac{2a}{2} + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right)$$

$$= \frac{-2}{h} \sin\left(a + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right) \quad \text{Ans.}$$

Q.3 Express the following: (Lahore Board 2009-2010)

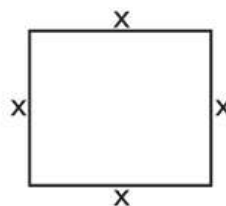
- The perimeter P of square as a function of its area A .
- The area A of a circle as a function of its circumference C .
- The volume V of a cube as a function of the area A of its base.

Solution:

(a) Let,

$$\text{Length of square} = x$$

$$\text{Width of square} = x$$



$$\text{Perimeter of a square} = P = x + x + x + x$$

$$P = 4x \quad \dots\dots\dots (1)$$

$$\text{Area of a square} = A = x \times x$$

$$A = x^2$$

$$x = \sqrt{A}$$

Put $x = \sqrt{A}$ in equation (1)

$$\boxed{P = 4\sqrt{A}}$$

Shows perimeter P of a square as a function of its area A .

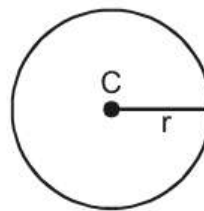


- (b) Let 'r' be the radius of the circle.

$$\text{Area of a circle} = A = \pi r^2 \quad \dots\dots\dots (1)$$

$$\text{Circumference of a circle} = C = 2\pi r$$

$$r = \frac{C}{2\pi}$$



$$\text{Put } r = \frac{C}{2\pi} \text{ in equation (1)}$$

$$A = \pi \left(\frac{C}{2\pi} \right)^2 = \frac{\pi C^2}{4\pi^2}$$

$$A = \frac{C^2}{4\pi}$$

Shows area A of a circle as a function of its circumference C.

- (c) Let x be the each side of cube. **(Gujranwala Board 2008)**

$$\text{Volume of cube} = V = x \times x \times x$$

$$V = x^3 \quad \dots\dots\dots (1)$$

$$\text{Area of base} = A = x \times x$$

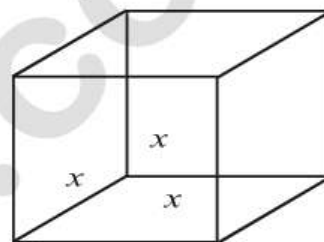
$$A = x^2$$

$$x = \sqrt{A}$$

$$\text{Put, } x = \sqrt{A} \text{ in equation (1)}$$

$$V = (\sqrt{A})^3$$

$$V = A^{3/2}$$



Shows volume V of a cube as a function of the area A of its base.

Q.4 Find the domain and the range of the function g defined below and sketch of graph of g.

(i) $g(x) = 2x - 5$

(ii) $g(x) = \sqrt{x^2 - 4}$

(iii) $g(x) = \sqrt{x+1}$ (Lhr.Board-2011)

(iv) $g(x) = |x - 3|$

(v) $g(x) = \begin{cases} 6x+7, & x \leq -2 \\ x-3, & x > -2 \end{cases}$

(vi) $g(x) = \begin{cases} x-1, & x < 3 \\ 2x+1, & 3 \leq x \end{cases}$

(vii) $g(x) = \frac{x^2 + 3x + 2}{x + 1}, x \neq -1$

(viii) $g(x) = \frac{x^2 - 16}{x - 4}, x \neq 4$

Solution:

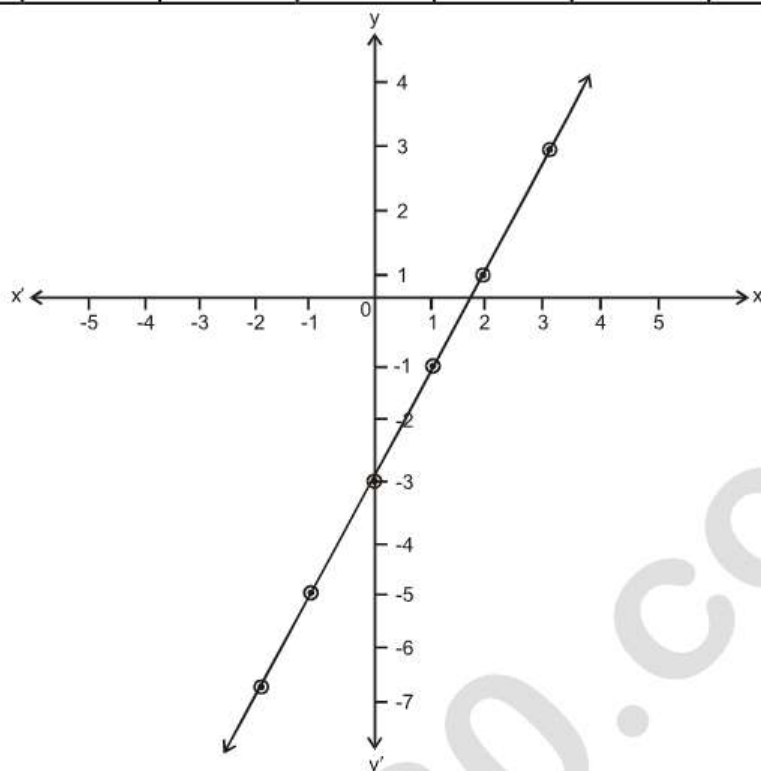
(i) $g(x) = 2x - 5$

Domain of $g(x)$ = Set of all real numbers

Range of $g(x)$ = Set of all real numbers



x	-3	-2	-1	0	1	2	3
$g(x) = 2x - 5$	-7	-9	-7	-5	-3	-1	1

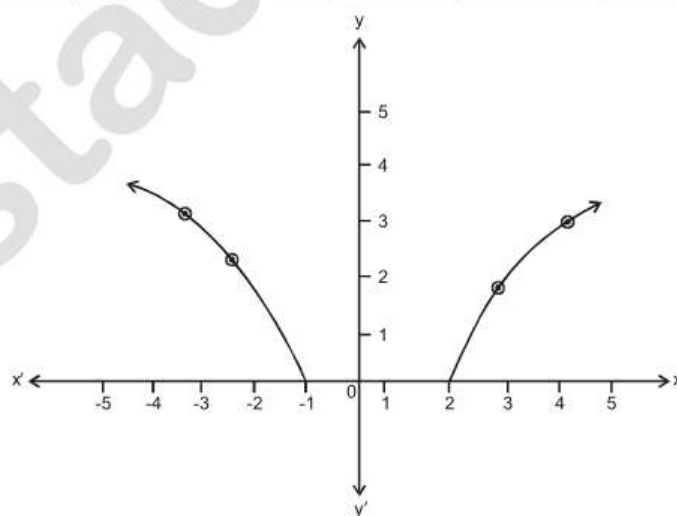


(ii) $g(x) = \sqrt{x^2 - 4}$ (Lahore Board 2008)

Domain of $g(x) = \mathbb{R} - (-2, 2)$

Range of $g(x) = [0, +\infty)$

x	-4	-3	-2	2	3	4
$g(x) = \sqrt{x^2 - 4}$	$2\sqrt{3}$	$\sqrt{5}$	0	0	$\sqrt{5}$	$2\sqrt{3}$

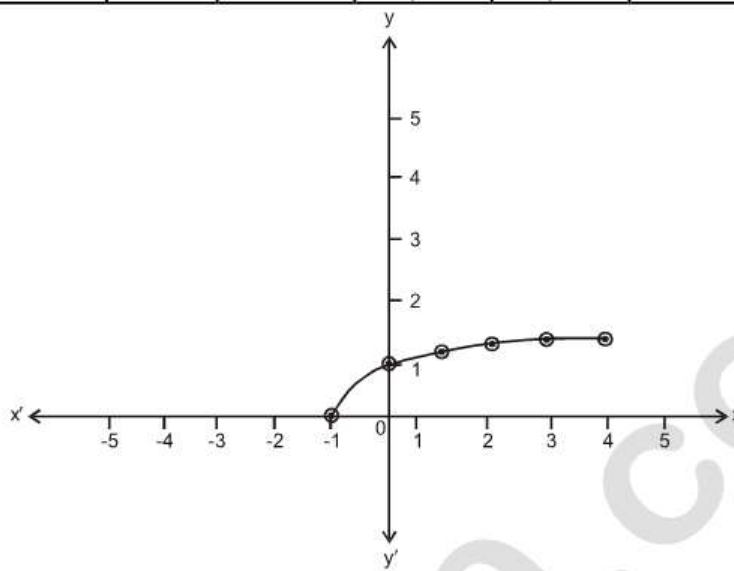


(iii) $g(x) = \sqrt{x+1}$

Domain of $g(x) = [-1, +\infty)$

Range of $g(x) = [0, +\infty)$

x	-1	0	1	2	3	4
$g(x) = \sqrt{x+1}$	0	1	$\sqrt{2}$	$\sqrt{3}$	2	$\sqrt{5}$

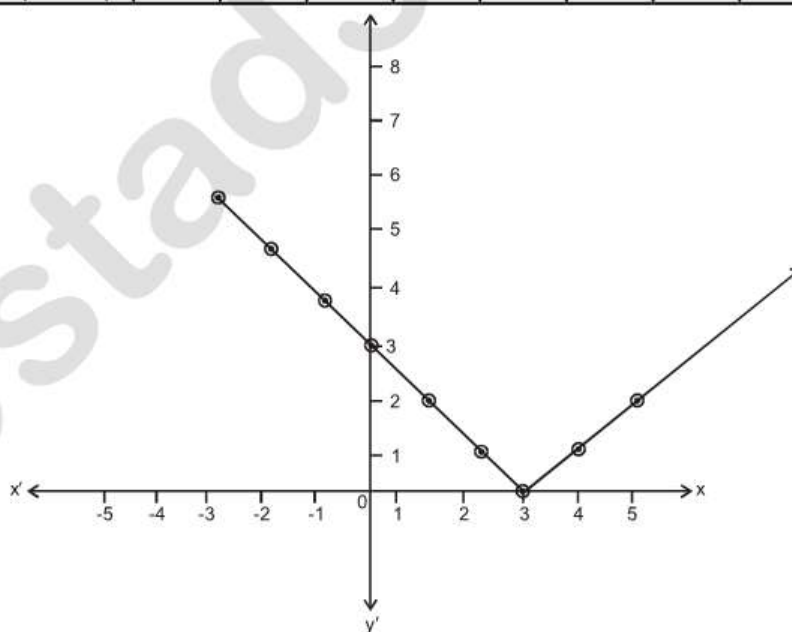


(iv) $g(x) = |x-3|$

Domain of $g(x) = (-\infty, +\infty)$

Range of $g(x) = [0, +\infty)$

x	-3	-2	-1	0	1	2	3	4	5
$g(x) = x-3 $	6	5	4	3	2	1	0	1	2



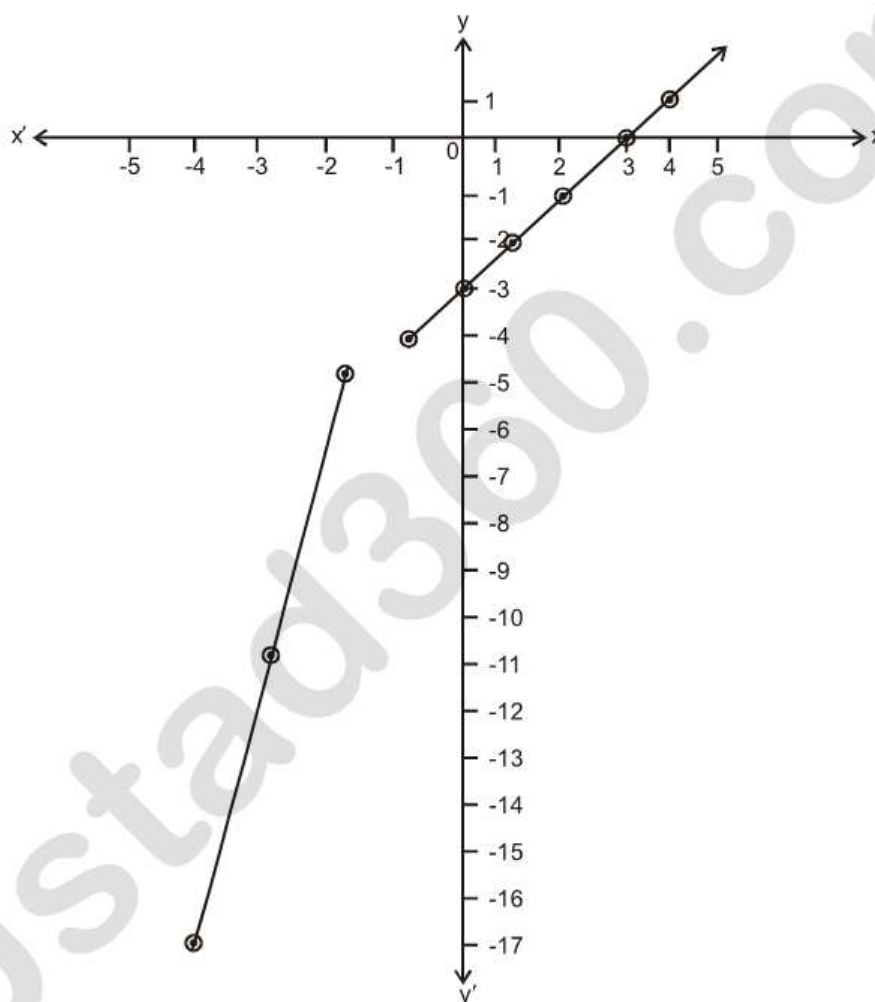
(v) $g(x) = \begin{cases} 6x + 7, & x \leq -2 \\ x - 3, & x > -2 \end{cases}$

Domain of $g(x) = (-\infty, +\infty)$

Range of $g(x) = (-\infty, +\infty)$

$x \leq -2$	-2	-3	-4	-5
$g(x) = 6x + 7$	-5	-11	-17	-23

$x > -2$	-1	0	1	2	3	3
$g(x) = x - 3$	-4	-3	-2	-1	0	1



(vi) $g(x) = \begin{cases} x - 1, & x < 3 \\ 2x + 1, & 3 \leq x \end{cases}$

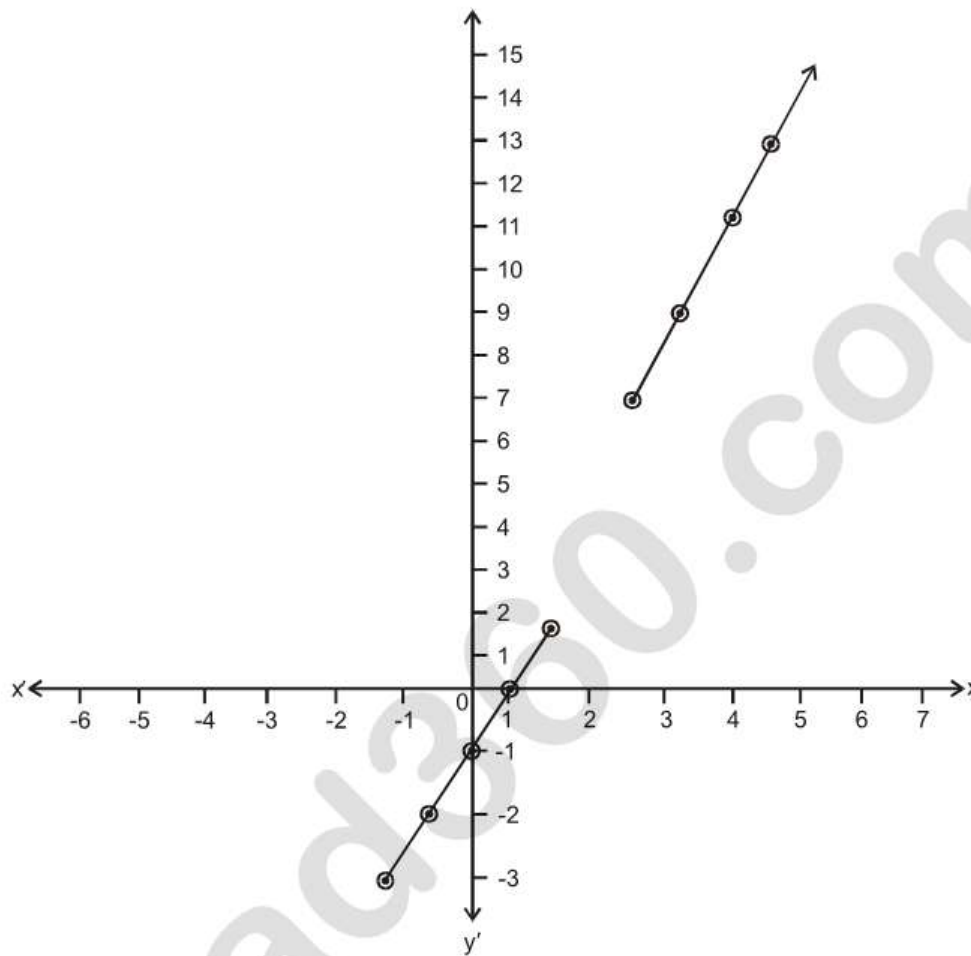
Domain of $g(x) = (-\infty, +\infty)$

Range of $g(x) = (-\infty, 2) \cup [7, +\infty)$



$x < 3$	-2	-1	0	1	2
$g(x) = x - 1$	-3	-2	-1	0	1

$x \geq 3$	3	4	5	6
$g(x) = 2x + 1$	7	9	11	13



(vii)
$$g(x) = \frac{x^2 + 3x + 2}{x + 1}, \quad x \neq -1$$

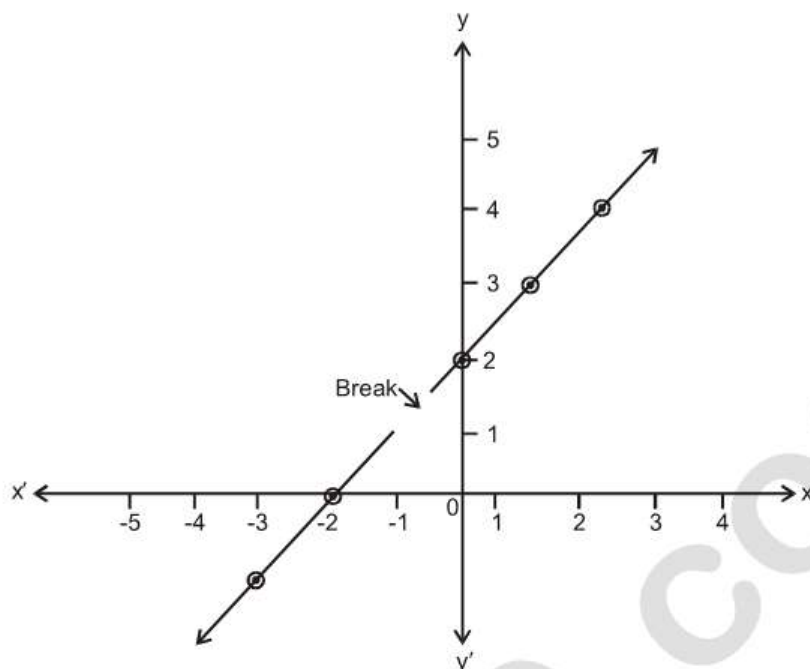
$$\begin{aligned}
 g(x) &= \frac{x^2 + 2x + x + 2}{x + 1} \\
 &= \frac{x(x + 2) + 1(x + 2)}{x + 1} \\
 &= \frac{(x + 2)(x + 1)}{x + 1} = x + 2
 \end{aligned}$$

Domain of $g(x) = \mathbb{R} - \{-1\}$

Range of $g(x) = \mathbb{R} - \{1\}$



x	-3	-2	0	1	2
$g(x) = x + 2$	-1	0	2	3	4



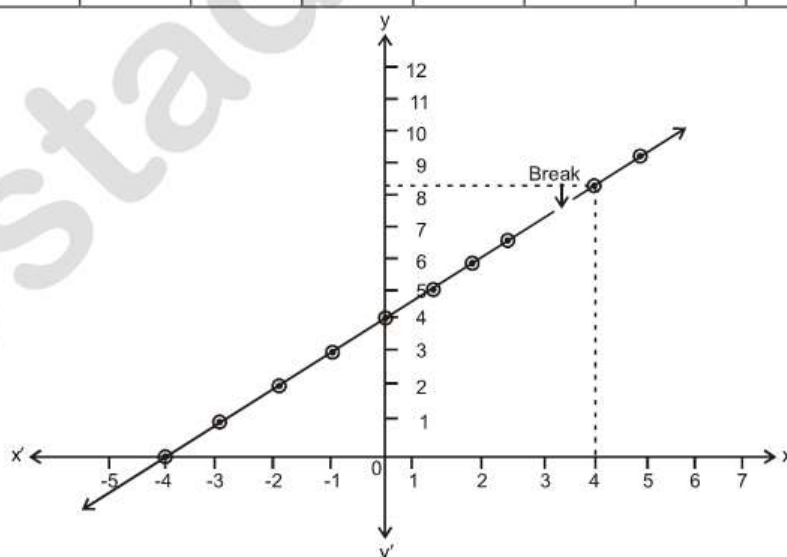
(viii)

$$g(x) = \frac{x^2 - 16}{x - 4}, \quad x \neq 4$$

$$= \frac{(x + 4)(x - 4)}{x - 4} = x + 4$$

Domain of $g(x) = \mathbb{R} - \{4\}$ Range of $g(x) = \mathbb{R} - \{8\}$

x	-4	-3	-2	-1	0	1	2	3	5	6
$g(x) = x + 4$	0	1	2	3	4	5	6	7	9	10



Q.5 Given $f(x) = x^3 - ax^2 + bx + 1$.

If $f(2) = -3$ and $f(-1) = 0$. Find the values of a and b .

Solution:

$$f(x) = x^3 - ax^2 + bx + 1 \quad \dots\dots\dots (1)$$

Put $x = 2$ in equation (1)

$$\begin{aligned} f(2) &= (2)^3 - a(2)^2 + b(2) + 1 \\ &= 8 - 4a + 2b + 1 \end{aligned}$$

$$f(2) = 9 - 4a + 2b$$

Put $x = -1$ in equation (1)

$$\begin{aligned} f(-1) &= (-1)^3 - a(-1)^2 + b(-1) + 1 \\ &= -1 - a - b + 1 \end{aligned}$$

$$f(-1) = -a - b$$

$$\text{Since } f(2) = -3$$

$$9 - 4a + 2b = -3$$

$$-4a + 2b = -3 - 9$$

$$-2(2a - b) = -12$$

$$2a - b = \frac{-12}{-2}$$

$$2a - b = 6 \quad \dots\dots\dots (2)$$

And

$$f(-1) = 0$$

$$-a - b = 0$$

$$-a = b$$

$$a = -b \quad \dots\dots\dots (3)$$

Put $a = -b$ in equation (2)

$$2(-b) - b = 6$$

$$-2b - b = 6$$

$$-3b = 6$$

$$b = \frac{6}{-3} = -2$$

Put $b = -2$ in equation (2)

$$a = -(-2)$$

$$a = 2$$

$\therefore$ $a = 2, b = -2$ Ans.



Q.6 A stone falls from a height of 60m on the ground, the height h after x second is approximately given by $h(x) = 40 - 10x^2$.

(i) What is the height of the stone when.

(a) $x = 1$ sec (b) $x = 1.5$ sec (c) $x = 1.7$ sec

(ii) When does the stone strike the ground?

Solution:

(i) $h(x) = 40 - 10x^2$ (1)

(a) Put $x = 1$ sec in equation (1)

$$\begin{aligned} h(1) &= 40 - 10(1)^2 \\ &= 40 - 10 \\ &= 30\text{m} \quad \text{Ans.} \end{aligned}$$

(b) Put $x = 1.5$ sec in equation (1)

$$\begin{aligned} h(1.5) &= 40 - 10(1.5)^2 \\ &= 40 - 10(2.25) \\ &= 40 - 22.5 \\ &= 17.5\text{m} \quad \text{Ans.} \end{aligned}$$

(c) Put $x = 1.7$ sec in equation (1)

$$\begin{aligned} h(1.7) &= 40 - 10(1.7)^2 \\ &= 40 - 10(2.89) \\ &= 40 - 28.9 \\ &= 11.1 \text{ m} \quad \text{Ans.} \end{aligned}$$

(ii) When then the stone strike the ground.

then $h(x) = 0$

$$0 = 40 - 10x^2$$

$$10x^2 = 40$$

$$x^2 = \frac{40}{10}$$

$$x^2 = 4$$

$$x = 2 \text{ sec} \quad \text{Ans.}$$

Q.7: Show that the Parametric equations.

(i) $x = at^2$, $y = 2at$ represent the equation of Parabola $y^2 = 4ax$

(ii) $x = a\cos\theta$, $y = b\sin\theta$ represent the equation of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(iii) $x = a\sec\theta$, $y = b\tan\theta$ represent the equation of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Solution:

(i) $x = at^2$ (i) , $y = 2at$ (ii)



From equation (ii)

$$t = \frac{y}{2a}$$

Putting it in (i)

$$x = a \left(\frac{y}{2a} \right)^2 = a \left(\frac{y^2}{4a^2} \right)$$

$$y^2 = 4ax \quad \text{Hence proved.}$$

$$(ii) \quad x = a \cos \theta$$

$$\frac{x}{a} = \cos \theta$$

Squaring on both sides

$$\frac{x^2}{a^2} = \cos^2 \theta \quad \dots (i)$$

Adding equation (i) & equation (ii)

$$\begin{aligned} \frac{x^2}{a^2} + \frac{y^2}{b^2} &= \cos^2 \theta + \sin^2 \theta \\ &= 1 \quad \text{Hence proved.} \end{aligned}$$

$$(iii) \quad x = a \sec \theta$$

$$\frac{x}{a} = \sec \theta$$

Squaring on both sides

$$\frac{x^2}{a^2} = \sec^2 \theta \quad \dots (i)$$

Subtracting equation (ii) from equation (i)

$$\begin{aligned} \frac{x^2}{a^2} - \frac{y^2}{b^2} &= \sec^2 \theta - \tan^2 \theta \\ &= 1 + \tan^2 \theta - \tan^2 \theta \\ &= 1 \quad \text{Hence proved.} \end{aligned}$$

Q.8 Prove the identities:

$$(i) \quad \sinh 2x = 2 \sinh x \cosh x$$

$$(ii) \quad \operatorname{sech}^2 x = 1 - \tanh^2 x$$

$$(iii) \quad \operatorname{cosech}^2 x = \coth^2 x - 1$$

(Lahore Board 2006)

Solution:

$$(i) \quad \sinh 2x = 2 \sinh x \cosh x$$

$$\text{R.H.S} = 2 \sinh x \cosh x$$

$$y = b \sin \theta$$

$$\frac{y}{b} = \sin \theta$$

Squaring on both sides

$$\frac{y^2}{b^2} = \sin^2 \theta \quad \dots (ii)$$

$$y = b \tan \theta$$

$$\frac{y}{b} = \tan \theta$$

Squaring on both sides

$$\frac{y^2}{b^2} = \tan^2 \theta \quad \dots (ii)$$



$$\begin{aligned}
 &= 2 \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^x + e^{-x}}{2} \right) \\
 &= \frac{e^{2x} - e^{-2x}}{2} \\
 &= \sinh 2x \\
 &= \text{L.H.S.} \quad \text{Hence proved.}
 \end{aligned}$$

(ii) **$\sec h^2 x = 1 - \tanh^2 x$**

$$\begin{aligned}
 \text{R.H.S} &= 1 - \tanh^2 x \\
 &= 1 - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2 \\
 &= 1 - \frac{(e^x - e^{-x})^2}{(e^x + e^{-x})^2} \\
 &= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2} \\
 &= \frac{e^{2x} + e^{-2x} + 2e^x \cdot e^{-x} - (e^{2x} + e^{-2x} - 2e^x \cdot e^{-x})}{(e^x + e^{-x})^2} \\
 &= \frac{e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 2}{(e^x + e^{-x})^2} \\
 &= \frac{4}{(e^x + e^{-x})^2} \\
 &= \left(\frac{2}{e^x + e^{-x}} \right)^2 \\
 &= (\operatorname{sech} x)^2 \\
 &= \operatorname{sech}^2 x \\
 &= \text{L.H.S} \quad \text{Hence proved}
 \end{aligned}$$

(iii) **$\operatorname{cosec} h^2 x = \coth^2 x - 1$**

$$\begin{aligned}
 \text{R.H.S} &= \coth^2 x - 1 \\
 &= \left(\frac{e^x + e^{-x}}{e^x - e^{-x}} \right)^2 - 1 \\
 &= \frac{(e^x + e^{-x})^2}{(e^x - e^{-x})^2} - 1 \\
 &= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x - e^{-x})^2} \\
 &= \frac{e^{2x} + e^{-2x} + 2e^x e^{-x} - (e^{2x} + e^{-2x} - 2e^x \cdot e^{-x})}{(e^x - e^{-x})^2}
 \end{aligned}$$



$$\begin{aligned}
 &= \frac{e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 2}{(e^x - e^{-x})^2} \\
 &= \frac{4}{(e^x + e^{-x})^2} \\
 &= \left(\frac{2}{e^x - e^{-x}} \right)^2 \\
 &= (\operatorname{cosech} x)^2 = \operatorname{cosech}^2 x \\
 &= \text{L.H.S} \quad \text{Hence proved}
 \end{aligned}$$

Q.9 Determine whether the given function f is even or odd:

(i) $f(x) = x^3 + x$

(ii) $f(x) = (x + 2)^2$

(iii) $f(x) = x\sqrt{x^2 + 5}$

(iv) $f(x) = \frac{x-1}{x+1}, x \neq -1$

(v) $f(x) = x^{2/3} + 6$

(vi) $f(x) = \frac{x^3 - x}{x^2 + 1}$

Solution:

(i) $f(x) = x^3 + x$

$$\begin{aligned}
 f(-x) &= (-x)^3 + (-x) \\
 &= -x^3 - x \\
 &= -(x^3 + x) \\
 &= -f(x)
 \end{aligned}$$

$\therefore f(x)$ is an odd function.

(ii) $f(x) = (x + 2)^2$

$$\begin{aligned}
 f(-x) &= (-x + 2)^2 \\
 &\neq \pm f(x)
 \end{aligned}$$

$\therefore f(x)$ is neither even nor odd function.

(iii) $f(x) = x\sqrt{x^2 + 5}$

$$\begin{aligned}
 f(-x) &= -x\sqrt{(-x)^2 + 5} \\
 &= -x\sqrt{x^2 + 5} \\
 &= -f(x)
 \end{aligned}$$

$\therefore f(x)$ is an odd function.

(iv) $f(x) = \frac{x-1}{x+1}, x \neq -1$

$$f(-x) = \frac{-x-1}{-x+1} = \frac{-(x+1)}{-(x-1)}$$



$$= \frac{x+1}{x-1} \neq \pm f(x)$$

$\therefore f(x)$ is neither even nor odd function.

(v) $f(x) = x^{2/3} + 6$

$$\begin{aligned} f(-x) &= (-x)^{2/3} + 6 \\ &= [(-x)^2]^{1/3} + 6 \\ &= (x^2)^{1/3} + 6 \\ &= x^{2/3} + 6 \\ &= f(x) \end{aligned}$$

$\therefore f(x)$ is an even function.

(vi) $f(x) = \frac{x^3 - x}{x^2 + 1}$

$$\begin{aligned} f(-x) &= \frac{(-x)^3 - (-x)}{(-x)^2 + 1} \\ &= \frac{-x^3 + x}{x^2 + 1} \\ &= \frac{-(x^3 - x)}{x^2 + 1} \\ &= -f(x) \end{aligned}$$

$\therefore f(x)$ is an odd function.

Composition of Functions:

Let f be a function from set X to set Y and g be a function from set Y to set Z . The composition of f and g is a function, denoted by $g \circ f$, from X to Z and is defined by.

$$(g \circ f)(x) = g(f(x)) = gf(x) \text{ for all } x \in X$$

Inverse of a Function:

Let f be one-one function from X onto Y . The inverse function of f , denoted by f^{-1} , is a function from Y onto X and is defined by.

$$x = f^{-1}(y), \forall y \in Y \text{ if and only if } y = f(x), \forall x \in X$$



EXERCISE 1.2

Q.1 The real valued functions f and g are defined below. Find

(a) $f \circ g(x)$ (b) $g \circ f(x)$ (c) $f \circ f(x)$ (d) $g \circ g(x)$

(i) $f(x) = 2x + 1$; $g(x) = \frac{3}{x-1}$, $x \neq 1$

(ii) $f(x) = \sqrt{x+1}$; $g(x) = \frac{1}{x^2}$, $x \neq 0$

(iii) $f(x) = \frac{1}{\sqrt{x-1}}$; $x \neq 1$; $g(x) = (x^2 + 1)^2$

(iv) $f(x) = 3x^4 - 2x^2$; $g(x) = \frac{2}{\sqrt{x}}$, $x \neq 0$



Solution:

$$(i) \quad f(x) = 2x + 1 \quad ; \quad g(x) = \frac{3}{x-1} \quad , \quad x \neq 1$$

$$\begin{aligned} (a) \quad fog(x) &= f(g(x)) \\ &= f\left(\frac{3}{x-1}\right) \\ &= 2\left(\frac{3}{x-1}\right) + 1 \\ &= \frac{6}{x-1} + 1 \\ &= \frac{6+x-1}{x-1} \\ &= \frac{x+5}{x-1} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} (b) \quad gof(x) &= g(f(x)) \\ &= g(2x+1) \\ &= \frac{3}{2x+1-1} = \frac{3}{2x} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} (c) \quad fof(x) &= f(f(x)) \\ &= f(2x+1) \\ &= 2(2x+1) + 1 \\ &= 4x + 2 + 1 \\ &= 4x + 3 \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} (d) \quad gog(x) &= g(g(x)) \\ &= g\left(\frac{3}{x-1}\right) \\ &= \frac{3}{\frac{3}{x-1}-1} \end{aligned}$$



$$\begin{aligned}
 &= \frac{3}{\frac{3 - (x - 1)}{x - 1}} \\
 &= \frac{3(x - 1)}{3 - x + 1} \\
 &= \frac{3(x - 1)}{4 - x} \quad \text{Ans.}
 \end{aligned}$$

(ii) $f(x) = \sqrt{x + 1}$; $g(x) = \frac{1}{x^2}$, $x \neq 0$

(a) $f \circ g(x) = f(g(x))$
 $= f\left(\frac{1}{x^2}\right)$
 $= \sqrt{\frac{1}{x^2} + 1}$
 $= \sqrt{\frac{1 + x^2}{x^2}} = \frac{\sqrt{1 + x^2}}{x} \quad \text{Ans.}$

(b) $g \circ f(x) = g(f(x))$
 $= g(\sqrt{x + 1})$
 $= \frac{1}{(\sqrt{x + 1})^2} = \frac{1}{x + 1} \quad \text{Ans.}$

(c) $f \circ f(x) = f(f(x))$
 $= f(\sqrt{x + 1})$
 $= \sqrt{\sqrt{x + 1} + 1} \quad \text{Ans.}$

(d) $g \circ g(x) = g(g(x))$
 $= g\left(\frac{1}{x^2}\right)$
 $= \frac{1}{\left(\frac{1}{x^2}\right)^2} = \frac{1}{\frac{1}{x^4}} = x^4 \quad \text{Ans.}$

(iii) $f(x) = \frac{1}{\sqrt{x - 1}}$; $x \neq 1$; $g(x) = (x^2 + 1)^2$

(a) $f \circ g(x) = f(g(x))$
 $= f((x^2 + 1)^2)$
 $= \frac{1}{\sqrt{(x^2 + 1)^2 - 1}}$



$$= \frac{1}{\sqrt{x^4 + 1 + 2x^2 - 1}}$$

$$= \frac{1}{\sqrt{x^2(x^2 + 2)}} = \frac{1}{x\sqrt{x^2 + 2}} \quad \text{Ans.}$$

(b) $\text{gof}(x) = g(f(x))$

$$= g\left(\frac{1}{\sqrt{x-1}}\right)$$

$$= \left[\left(\frac{1}{\sqrt{x-1}}\right)^2 + 1\right]^2$$

$$= \left(\frac{1}{x-1} + 1\right)^2 = \left(\frac{1+x-1}{x-1}\right)^2$$

$$= \left(\frac{x}{x-1}\right)^2 \quad \text{Ans.}$$

(c) $\text{fof}(x) = f(f(x))$

$$= f\left(\frac{1}{\sqrt{x-1}}\right)$$

$$= \frac{1}{\sqrt{\sqrt{\frac{1}{\sqrt{x-1}}} - 1}}$$

$$= \frac{1}{\sqrt{\frac{1 - \sqrt{x-1}}{\sqrt{x-1}}}} = \sqrt{\frac{\sqrt{x-1}}{1 - \sqrt{x-1}}} \quad \text{Ans.}$$

(d) $\text{gog}(x) = g(g(x))$

$$= g((x^2 + 1)^2)$$

$$= [\{(x^2 + 1)^2\}^2 + 1]^2$$

$$= [(x^2 + 1)^4 + 1]^2 \quad \text{Ans.}$$

(iv) $f(x) = 3x^4 - 2x^2$; $g(x) = \frac{2}{\sqrt{x}}$, $x \neq 0$

(a) $\text{fog}(x) = f(g(x))$

$$= f\left(\frac{2}{\sqrt{x}}\right)$$

$$= 3\left(\frac{2}{\sqrt{x}}\right)^4 - 2\left(\frac{2}{\sqrt{x}}\right)^2$$



$$\begin{aligned}
 &= 3\left(\frac{16}{x^2}\right) - 2\left(\frac{4}{x}\right) \\
 &= \frac{48}{x^2} - \frac{8}{x} \\
 &= \frac{48 - 8x}{x^2} \\
 &= \frac{8(6 - x)}{x^2} \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \text{gof}(x) &= g(f(x)) \\
 &= g(3x^4 - 2x^2) \\
 &= \frac{2}{\sqrt{3x^4 - 2x^2}} \\
 &= \frac{2}{\sqrt{x^2(3x^2 - 2)}} = \frac{2}{x\sqrt{3x^2 - 2}} \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad \text{fof}(x) &= f(f(x)) \\
 &= f(3x^4 - 2x^2) \\
 &= 3(3x^4 - 2x^2)^4 - 2(3x^4 - 2x^2)^2 \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \text{gog}(x) &= g(g(x)) \\
 &= g\left(\frac{2}{\sqrt{x}}\right) \\
 &= \frac{2}{\sqrt{2/\sqrt{x}}} \\
 &= 2\sqrt{\frac{\sqrt{x}}{2}} \\
 &= \sqrt{2} \times \sqrt{2} \frac{\sqrt{\sqrt{x}}}{\sqrt{2}} \\
 &= \sqrt{2} \sqrt{x} \quad \text{Ans.}
 \end{aligned}$$

Q.2 For the real valued function, f defined below, find:

(a) $f^{-1}(x)$

(b) $f^{-1}(-1)$ and verify $f(f^{-1}(x)) = f^{-1}(f(x)) = x$

(i) $f(x) = -2x + 8$ (Lahore Board 2007, 2009)

(ii) $f(x) = 3x^3 + 7$

(iii) $f(x) = (-x + 9)^3$

(iv) $f(x) = \frac{2x+1}{x-1}, x > 1$



Solution:

(i) $f(x) = -2x + 8$

(a) Since $y = f(x)$
 $\Rightarrow x = f^{-1}(y)$

Now,

$$f(x) = -2x + 8$$

$$y = -2x + 8$$

$$2x = 8 - y$$

$$x = \frac{8 - y}{2}$$

$$f^{-1}(y) = \frac{8 - y}{2}$$

Replacing y by x

$$f^{-1}(x) = \frac{8 - x}{2}$$

Replacing y by x.

$$\boxed{f^{-1}(x) = \frac{8 - x}{2}}$$

(b) Put, $x = -1$

$$f^{-1}(-1) = \frac{8 - (-1)}{2} = \frac{8 + 1}{2} = \frac{9}{2}$$

$$\begin{aligned} f(f^{-1}(x)) &= f\left(\frac{8 - x}{2}\right) \\ &= -2\left(\frac{8 - x}{2}\right) + 8 \\ &= -8 + x + 8 \\ &= x \end{aligned}$$

$$\begin{aligned} f^{-1}(f(x)) &= f^{-1}(-2x + 8) \\ &= \frac{8 - (-2x + 8)}{2} \\ &= \frac{8 + 2x - 8}{2} \\ &= \frac{2x}{2} = x \end{aligned}$$

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x \quad \text{Hence proved.}$$



$$(ii) \quad f(x) = 3x^3 + 7$$

$$(a) \quad \text{Since } y = f(x) \\ \Rightarrow x = f^{-1}(y)$$

Now

$$f(x) = 3x^3 + 7$$

$$y = 3x^3 + 7$$

$$3x^3 = y - 7$$

$$x^3 = \frac{y-7}{3}$$

$$x = \left(\frac{y-7}{3} \right)^{\frac{1}{3}}$$

$$f^{-1}(y) = \left(\frac{y-7}{3} \right)^{\frac{1}{3}}$$

Replacing y by x

$$f^{-1}(x) = \left(\frac{x-7}{3} \right)^{\frac{1}{3}}$$

$$(b) \quad \text{Put } x = -1$$

$$f^{-1}(-1) = \left(\frac{-1-7}{3} \right)^{\frac{1}{3}}$$

$$= \left(\frac{-8}{3} \right)^{\frac{1}{3}}$$

$$f(f^{-1}(x)) = f \left[\left(\frac{x-7}{3} \right)^{\frac{1}{3}} \right]$$

$$= 3 \left[\left(\frac{x-7}{3} \right)^{\frac{1}{3}} \right]^3 + 7$$

$$= 3 \left(\frac{x-7}{3} \right) + 7$$

$$= x - 7 + 7 = x$$

$$f^{-1}(f(x)) = f^{-1}(3x^3 + 7)$$

$$= \left(\frac{3x^3 + 7 - 7}{3} \right)^{\frac{1}{3}}$$



$$= \left(\frac{3x^3}{3} \right)^{\frac{1}{3}}$$

$$= (x^3)^{\frac{1}{3}} = x$$

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x \quad \text{Hence proved.}$$

(iii) $f(x) = (-x + 9)^3$

(a) Since $y = f(x)$
 $x = f^{-1}(y)$

Now

$$f(x) = (-x + 9)^3$$

$$y = (-x + 9)^3$$

$$y^{\frac{1}{3}} = -x + 9$$

$$x = 9 - y^{\frac{1}{3}}$$

Replacing y by x

$$f^{-1}(x) = 9 - x^{\frac{1}{3}}$$

(b) Put $x = -1$

$$f^{-1}(-1) = 9 - (-1)^{\frac{1}{3}}$$

$$f(f^{-1}(x)) = f(9 - x^{\frac{1}{3}})$$

$$= [-(9 - x^{\frac{1}{3}}) + 9]^3$$

$$= (-9 + x^{\frac{1}{3}} + 9)^3$$

$$= \left(x^{\frac{1}{3}} \right)^3 = x$$

$$f^{-1}(f(x)) = f^{-1}((-x + 9)^3)$$

$$= 9 - [(-x + 9)^3]^{\frac{1}{3}}$$

$$= 9 - (-x + 9)$$

$$= 9 + x - 9$$

$$= x$$

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x \quad \text{Hence proved.}$$



$$(iv) \quad f(x) = \frac{2x+1}{x-1}, \quad x > 1$$

$$(a) \quad \text{Since } y = f(x) \\ x = f^{-1}(y)$$

Now

$$f(x) = \frac{2x+1}{x-1}$$

$$y = \frac{2x+1}{x-1}$$

$$y(x-1) = 2x+1$$

$$yx - y = 2x + 1$$

$$yx - 2x = 1 + y$$

$$x(y-2) = y+1$$

$$x = \frac{y+1}{y-2}$$

$$f^{-1}(y) = \frac{y+1}{y-2}$$

Replacing y by x

$$f^{-1}(x) = \frac{x+1}{x-2}$$

$$(b) \quad \text{Put } x = -1$$

$$f^{-1}(-1) = \frac{-1+1}{-1-2}$$

$$= \frac{0}{-3} = 0$$

$$f(f^{-1}(x)) = f\left(\frac{x+1}{x-2}\right) \\ = \frac{2\left(\frac{x+1}{x-2}\right)+1}{\frac{x+1}{x-2}-1}$$

$$= \frac{\frac{2(x+1) + (x-2)}{x-2}}{\frac{x+1 - (x-2)}{x-2}}$$



$$\begin{aligned}
 &= \frac{2x + 2 + x - 2}{x + 1 - x + 2} \\
 &= \frac{3x}{3} = x \\
 f^{-1}(f(x)) &= f^{-1}\left(\frac{2x+1}{x-1}\right) \\
 &= \frac{\frac{2x+1}{x-1} + 1}{\frac{2x+1}{x-1} - 2} \\
 &= \frac{\frac{2x+1+x-1}{x-1}}{\frac{2x+1-2(x-1)}{x-1}} \\
 &= \frac{3x}{2x+1-2x+2} \\
 &= \frac{3x}{3} = x
 \end{aligned}$$

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x \quad \text{Hence proved.}$$

Q.3 Without finding the inverse, state the domain and range of f^{-1} .

$$\begin{aligned}
 \text{(i)} \quad f(x) &= \sqrt{x+2} & \text{(ii)} \quad f(x) &= \frac{x-1}{x-4}, x \neq 4 \\
 \text{(iii)} \quad f(x) &= \frac{1}{x+3}, x \neq -3 & \text{(iv)} \quad f(x) &= (x-5)^2, x \geq 5
 \end{aligned}$$

Solution:

$$\begin{aligned}
 \text{(i)} \quad f(x) &= \sqrt{x+2} \\
 \text{Domain of } f(x) &= [-2, +\infty) \\
 \text{Range of } f(x) &= [0, +\infty) \\
 \text{Domain of } f^{-1}(x) &= \text{Range of } f(x) = [0, +\infty) \\
 \text{Range of } f^{-1}(x) &= \text{Domain of } f(x) = [-2, +\infty)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad f(x) &= \frac{x-1}{x-4}, x \neq 4 \\
 \text{Domain of } f(x) &= \mathbb{R} - \{4\} \\
 \text{Range of } f(x) &= \mathbb{R} - \{1\} \\
 \text{Domain of } f^{-1}(x) &= \text{Range of } f(x) = \mathbb{R} - \{1\} \\
 \text{Range of } f^{-1}(x) &= \text{Domain of } f(x) = \mathbb{R} - \{4\}
 \end{aligned}$$



(iii) $f(x) = \frac{1}{x+3}, x \neq -3$

Domain of $f(x)$ = $\mathbb{R} - \{-3\}$

Range of $f(x)$ = $\mathbb{R} - \{0\}$

Domain of $f^{-1}(x)$ = Range of $f(x)$ = $\mathbb{R} - \{0\}$

Range of $f^{-1}(x)$ = Domain of $f(x)$ = $\mathbb{R} - \{-3\}$

(iv) $f(x) = (x-5)^2, x \geq 5$ (Gujranwala Board 2007)

Domain of $f(x)$ = $[5, +\infty)$

Range of $f(x)$ = $[0, +\infty)$

Domain of $f^{-1}(x)$ = Range of $f(x)$ = $[0, +\infty)$

Range of $f^{-1}(x)$ = Domain of $f(x)$ = $[5, +\infty)$

Limit of a Function:

Let a function $f(x)$ be defined in an open interval near the number 'a' (need not at a) if, as x approaches 'a' from both left and right side of 'a', $f(x)$ approaches a specific number 'L' then 'L', is called the limit of $f(x)$ as x approaches a symbolically it is written as.

$$\lim_{x \rightarrow a} f(x) = L \text{ read as "Limit of } f(x) \text{ as } x \rightarrow a, \text{ is } L"$$

Theorems on Limits of Functions:

Let f and g be two functions, for which $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then

Theorem 1: The limit of the sum of two functions is equal to the sum of their limits.

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) + g(x)] &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) \\ &= L + M \end{aligned}$$

Theorem 2: The limit of the difference of two functions is equal to the difference of their limits.

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) - g(x)] &= \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) \\ &= L - M \end{aligned}$$

Theorem 3: If K is any real numbers, then.

$$\lim_{x \rightarrow a} [kf(x)] = K \lim_{x \rightarrow a} f(x) = kL$$

Theorem 4: The limit of the product of the functions is equal to the product of their limits.

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = [\lim_{x \rightarrow a} f(x)] [\lim_{x \rightarrow a} g(x)] = LM$$

Theorem 5: The limit of the quotient of the functions is equal to the quotient of their limits provided the limit of the denominator is non-zero.



$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{M}, \quad g(x) \neq 0, M \neq 0$$

Theorem 6: Limit of $[f(x)]^n$, where n is an integer.

$$\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n = L^n$$

The Sandwich Theorem:

Let f , g and h be functions such that $f(x) \leq g(x) \leq h(x)$ for all number x in some open interval containing “ C ”, except possibly at C itself.

If, $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} h(x) = L$, then $\lim_{x \rightarrow c} g(x) = L$.

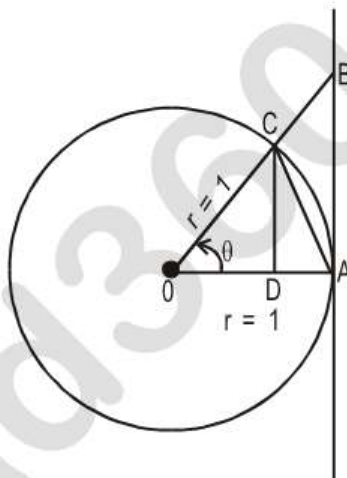
Prove that

If θ is measured in radian, then

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

Proof:

Take θ a positive acute central angle of a circle with radius $r = 1$. OAB represents the sector of the circle.



$$|OA| = |OC| = 1 \quad (\text{radii of unit circle})$$

From right angle $\triangle ODC$

$$\sin \theta = \frac{|DC|}{|OC|} = |DC| \quad (\because |OC| = 1)$$

From right angle $\triangle OAB$

$$\tan \theta = \frac{|AB|}{|OA|} = |AB| \quad (\because |OA| = 1)$$

In terms of θ , the areas are expressed as

$$\text{Area of } \triangle OAC = \frac{1}{2} |OA| |CD| = \frac{1}{2} (1) \sin \theta = \frac{1}{2} \sin \theta$$



$$\text{Area of sector OAC} = \frac{1}{2} r^2 \theta = \frac{1}{2} (1)(\theta) = \frac{1}{2} \theta$$

$$\text{Area of } \triangle OAB = \frac{1}{2} |OA| |AB| = \frac{1}{2} (1) \tan \theta = \frac{1}{2} \tan \theta$$

From figure

Area of $\triangle OAB >$ Area of sector OAC $>$ Area of $\triangle OAC$

$$\frac{1}{2} \tan \theta > \frac{1}{2} \theta > \frac{1}{2} \sin \theta$$

$$\frac{1}{2} \frac{\sin \theta}{\cos \theta} > \frac{\theta}{2} > \frac{\sin \theta}{2}$$

As $\sin \theta$ is positive, so on division by $\frac{1}{2} \sin \theta$, we get.

$$\frac{1}{\cos \theta} > \frac{\theta}{\sin \theta} > 1 \quad (0 < \theta < \pi/2)$$

i.e.

$$\cos \theta < \frac{\sin \theta}{\theta} < 1$$

When, $\theta \rightarrow 0$, $\cos \theta \rightarrow 1$

Since $\frac{\sin \theta}{\theta}$ is sandwiched between 1 and a quantity approaching 1 itself.

So by the sandwich theorem it must also approach 1.

i.e.

$$\boxed{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1}$$

Theorem: Prove that

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$$

Proof:

Taking

$$\begin{aligned} \left(1 + \frac{1}{n}\right)^n &= 1 + n\left(\frac{1}{n}\right) + \frac{n(n-1)}{2!} \left(\frac{1}{n}\right)^2 + \frac{n(n-1)(n-2)}{3!} \left(\frac{1}{n}\right)^3 + \dots \\ &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \dots \end{aligned}$$

Taking $\lim_{n \rightarrow +\infty}$ on both sides.

$$\begin{aligned} \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \dots \\ &= 1 + 1 + 0.5 + 0.166667 + 0.0416667 + \dots \end{aligned}$$



$$= 2.718281 \dots\dots\dots$$

As approximate value of e is $= 2.718281$

$$\therefore \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$$

Deduction:

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$$

We know that.

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$$

Put $x = \frac{1}{n}$ then $\frac{1}{x} = n$

As $n \rightarrow +\infty$, $x \rightarrow 0$

$$\therefore \lim_{n \rightarrow +\infty} (1 + x)^{1/x} = e$$

Theorem:

Prove that:

$$\lim_{x \rightarrow a} \frac{a^x - 1}{x} = \log_e a$$

Proof:

Taking,

$$\lim_{x \rightarrow a} \frac{a^x - 1}{x}$$

Let $a^x - 1 = y$

$$a^x = 1 + y$$

$$x = \log_a (1 + y)$$

As, $x \rightarrow a$, $y \rightarrow 0$

$$\lim_{x \rightarrow a} \frac{a^x - 1}{x} = \lim_{y \rightarrow 0} \frac{y}{\log_a (1 + y)}$$

$$= \lim_{y \rightarrow 0} \frac{1}{\frac{1}{y} \log_a (1 + y)}$$

$$= \frac{1}{\log_a e}$$

$$= \log_e a$$

$$= \lim_{y \rightarrow 0} \frac{1}{\log_a (1 + y)^{\frac{1}{y}}}$$

$$\therefore \lim_{y \rightarrow 0} (1 + y)^{1/y} = e$$



Deduction

$$\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = \log_e e = 1$$

We know that

$$\lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log_e a$$

Put $a = e$

$$\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = \log_e e = 1$$

Important results to remember

- (i) $\lim_{x \rightarrow +\infty} (e^x) = \infty$ (ii) $\lim_{x \rightarrow -\infty} (e^x) = \lim_{x \rightarrow -\infty} \left(\frac{1}{e^{-x}} \right) = 0$
- (iii) $\lim_{x \rightarrow \pm\infty} \left(\frac{a}{x} \right) = 0$, where a is any real number.



EXERCISE 1.3

Q.1 Evaluate each limit by using theorems of limits.

(i) $\lim_{x \rightarrow 3} (2x + 4)$

(ii) $\lim_{x \rightarrow 1} (3x^2 - 2x + 4)$

(iii) $\lim_{x \rightarrow 3} \sqrt{x^2 + x + 4}$

(iv) $\lim_{x \rightarrow 2} x\sqrt{x^2 - 4}$

(v) $\lim_{x \rightarrow 2} (\sqrt{x^3 + 1} - \sqrt{x^2 + 5})$

(iv) $\lim_{x \rightarrow 2} \frac{2x^3 + 5x}{3x - 2}$

Solution:

$$\begin{aligned} \text{(i)} \quad \lim_{x \rightarrow 3} (2x + 4) &= \lim_{x \rightarrow 3} (2x) + \lim_{x \rightarrow 3} (4) \\ &= 2 \lim_{x \rightarrow 3} x + 4 \\ &= 2(3) + 4 = 6 + 4 = 10 \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \lim_{x \rightarrow 1} (3x^2 - 2x + 4) &= \lim_{x \rightarrow 1} (3x^2) - \lim_{x \rightarrow 1} (2x) + \lim_{x \rightarrow 1} (4) \\ &= 3 \lim_{x \rightarrow 1} x^2 - 2 \lim_{x \rightarrow 1} x + 4 \\ &= 3(1)^2 - 2(1) + 4 \\ &= 3 - 2 + 4 \\ &= 5 \quad \text{Ans.} \end{aligned}$$

$$\text{(iii)} \quad \lim_{x \rightarrow 3} \sqrt{x^2 + x + 4} = [\lim_{x \rightarrow 3} (x^2 + x + 4)]^{1/2}$$



$$\begin{aligned}
 &= [\lim_{x \rightarrow 3} x^2 + \lim_{x \rightarrow 3} x + \lim_{x \rightarrow 3} 4]^{1/2} \\
 &= (3^2 + 3 + 4)^{1/2} \\
 &= (9 + 7)^{1/2} = (16)^{1/2} = (4^2)^{1/2} = 4 \text{ Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad \lim_{x \rightarrow 2} x\sqrt{x^2 - 4} &= [\lim_{x \rightarrow 2} (x)] [\lim_{x \rightarrow 2} (x^2 - 4)^{1/2}] \\
 &= 2 [\lim_{x \rightarrow 2} (x^2 - 4)]^{1/2} \\
 &= 2 [\lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} 4]^{1/2} \\
 &= 2 (4 - 4)^{1/2} \\
 &= 2(0)^{1/2} \\
 &= 0 \text{ Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \lim_{x \rightarrow 2} (\sqrt{x^3 + 1} - \sqrt{x^2 + 5}) &= \lim_{x \rightarrow 2} (x^3 + 1)^{1/2} - \lim_{x \rightarrow 2} (x^2 + 5)^{1/2} \\
 &= [\lim_{x \rightarrow 2} (x^3 + 1)]^{1/2} - [\lim_{x \rightarrow 2} (x^2 + 5)]^{1/2} \\
 &= [\lim_{x \rightarrow 2} x^3 + \lim_{x \rightarrow 2} 1]^{1/2} - [\lim_{x \rightarrow 2} x^2 + \lim_{x \rightarrow 2} 5]^{1/2} \\
 &= (8 + 1)^{1/2} - (4 + 5)^{1/2} \\
 &= (9)^{1/2} - (9)^{1/2} = (3^2)^{1/2} - (3^2)^{1/2} = 3 - 3 \\
 &= 0 \text{ Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi)} \quad \lim_{x \rightarrow -2} \frac{2x^3 + 5x}{3x - 2} &= \frac{\lim_{x \rightarrow -2} (2x^3 + 5x)}{\lim_{x \rightarrow -2} (3x - 2)} \\
 &= \frac{2 \lim_{x \rightarrow -2} x^3 + 5 \lim_{x \rightarrow -2} x}{3 \lim_{x \rightarrow -2} x - \lim_{x \rightarrow -2} 2} \\
 &= \frac{2(-2)^3 + 5(-2)}{3(-2) - 2} \\
 &= \frac{2(-8) - 10}{-6 - 2} = \frac{-16 - 10}{-8} = \frac{-26}{-8} = \frac{13}{4} \text{ Ans}
 \end{aligned}$$

Q.2 Evaluate each limit by using algebraic techniques.

$$\text{(i)} \quad \lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1}$$

$$\text{(ii)} \quad \lim_{x \rightarrow 1} \left(\frac{3x^3 + 4x}{x^2 + x} \right)$$

$$\text{(iii)} \quad \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 + x - 6}$$

$$\text{(iv)} \quad \lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 3x - 1}{x^3 - x} \quad (\text{Lhr. Board 2009})$$

$$\text{(v)} \quad \lim_{x \rightarrow -1} \left(\frac{x^3 + x^2}{x^2 - 1} \right)$$

$$\text{(vi)} \quad \lim_{x \rightarrow 4} \frac{2x^2 - 32}{x^3 - 4x^2}$$



$$(vii) \quad \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} \quad (\text{Lhr. Board 2006})$$

$$(viii) \quad \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \quad (\text{Lhr. Board 2004})$$

$$(ix) \quad \lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m}$$

Solution:

$$(i) \quad \lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1} &= \lim_{x \rightarrow -1} \frac{x(x^2 - 1)}{x + 1} \\ &= \lim_{x \rightarrow -1} \frac{x(x+1)(x-1)}{x+1} \\ &= \lim_{x \rightarrow -1} x(x-1) \\ &= -1(-1-1) \\ &= -1(-2) = 2 \quad \text{Ans.} \end{aligned}$$

$$(ii) \quad \lim_{x \rightarrow 1} \left(\frac{3x^3 + 4x}{x^2 + x} \right) = \frac{3(1)^3 + 4(1)}{(1)^2 + 1} = \frac{3+4}{2} = \frac{7}{2} \quad \text{Ans.}$$

$$(iii) \quad \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 + x - 6} \quad \left(\frac{0}{0}\right) \text{ form} \quad (\text{Gujranwala 2007, Lahore Board 2008})$$

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 + x - 6} &= \lim_{x \rightarrow 2} \frac{(x)^3 - (2)^3}{x^2 + 3x - 2x - 6} \\ &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{x(x+3) - 2(x+3)} \quad [\because a^3 - b^3 = (a-b)(a^2 + ab + b^2)] \\ &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{(x+3)(x-2)} \\ &= \frac{(2)^2 + 2(2) + 4}{2+3} = \frac{4+4+4}{5} = \frac{12}{5} \quad \text{Ans.} \end{aligned}$$

$$(iv) \quad \lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 3x - 1}{x^3 - x} \quad \left(\frac{0}{0}\right) \text{ form} \quad (\text{Lahore Board 2009})$$

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 3x - 1}{x^3 - x} &= \lim_{x \rightarrow 1} \frac{(x-1)^3}{x(x^2 - 1)} \\ &= \lim_{x \rightarrow 1} \frac{(x-1)^3}{x(x+1)(x-1)} \end{aligned}$$



$$\begin{aligned}
 &= \lim_{x \rightarrow 1} \frac{(x-1)^2}{x(x+1)} \\
 &= \frac{(1-1)^2}{1(1+1)} = \frac{0}{2} = 0 \quad \text{Ans.}
 \end{aligned}$$

(v) $\lim_{x \rightarrow -1} \left(\frac{x^3 + x^2}{x^2 - 1} \right) \left(\frac{0}{0} \right) \text{form}$

$$\begin{aligned}
 \lim_{x \rightarrow -1} \left(\frac{x^3 + x^2}{x^2 - 1} \right) &= \lim_{x \rightarrow -1} \frac{x^2(x+1)}{(x+1)(x-1)} \\
 &= \lim_{x \rightarrow -1} \frac{x^2}{x-1} \\
 &= \frac{(-1)^2}{-1-1} = \frac{-1}{2} \quad \text{Ans.}
 \end{aligned}$$

(vi) $\lim_{x \rightarrow 4} \frac{2x^2 - 32}{x^3 - 4x^2} \left(\frac{0}{0} \right) \text{form}$

$$\begin{aligned}
 \lim_{x \rightarrow 4} \frac{2x^2 - 32}{x^3 - 4x^2} &= \lim_{x \rightarrow 4} \frac{2(x^2 - 16)}{x^2(x-4)} \\
 &= \lim_{x \rightarrow 4} \frac{2(x+4)(x-4)}{x^2(x-4)} \\
 &= \lim_{x \rightarrow 4} \frac{2(x+4)}{x^2} \\
 &= \frac{2(4+4)}{(4)^2} = \frac{2(8)}{16} \\
 &= \frac{16}{16} = 1 \quad \text{Ans.}
 \end{aligned}$$

(vii) $\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} \left(\frac{0}{0} \right) \text{form} \quad (\text{Guj. Board 2006})$

$$\begin{aligned}
 \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} &= \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} \times \frac{\sqrt{x} + \sqrt{2}}{\sqrt{x} + \sqrt{2}} \\
 &= \lim_{x \rightarrow 2} \frac{(\sqrt{x})^2 - (\sqrt{2})^2}{(x - 2)(\sqrt{x} + \sqrt{2})} \\
 &= \lim_{x \rightarrow 2} \frac{x - 2}{(x - 2)(\sqrt{x} + \sqrt{2})} \\
 &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x} + \sqrt{2}} = \frac{1}{\sqrt{2} + \sqrt{2}} = \frac{1}{2\sqrt{2}} \quad \text{Ans.}
 \end{aligned}$$



(viii) $\lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \quad \left(\frac{0}{0}\right) \text{ form} \quad (\text{Lahore Board 2006})$

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h})^2 - (\sqrt{x})^2}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \frac{1}{\sqrt{x+0} + \sqrt{x}} = \frac{1}{\sqrt{x} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}} \quad \text{Ans.} \end{aligned}$$

(ix) $\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} \quad \left(\frac{0}{0}\right) \text{ form}$

We know that:

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}, \quad \text{where } n \text{ is an integer and } a > 0$$

Now,

$$\begin{aligned} \lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} &= \lim_{x \rightarrow a} \frac{\frac{x^n - a^n}{x - a}}{\frac{x^m - a^m}{x - a}} \\ &= \frac{na^{n-1}}{ma^{m-1}} = \frac{n}{m} a^{n-1-m+1} = \frac{n}{m} a^{n-m} \quad \text{Ans.} \end{aligned}$$

Q.3 Evaluate the following limits:

(i) $\lim_{x \rightarrow 0} \frac{\sin 7x}{x}$

(ii) $\lim_{x \rightarrow 0} \frac{\sin x^0}{x} \quad (\text{L.B 2003})$

(iii) $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta} \quad (\text{L.B 2009 (s)})$

(iv) $\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$

(v) $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$

(vi) $\lim_{x \rightarrow 0} \frac{x}{\tan x}$

(vii) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

(viii) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x} \quad (\text{L.B 2009})$



- (ix) $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta}$ (L.B 2007) (x) $\lim_{x \rightarrow 0} \frac{\sec x - \cos x}{x}$
- (xi) $\lim_{\theta \rightarrow 0} \frac{1 - \cos p\theta}{1 - \cos q\theta}$ (L.B 2004,06) (G.B 2005, 2006)
- (xii) $\lim_{\theta \rightarrow 0} \frac{\tan \theta - \sin \theta}{\sin^3 \theta}$ (L.B 2003, 2004) (G.B 2005)

Solution:

(i) $\lim_{x \rightarrow 0} \frac{\sin 7x}{x}$ $\left(\frac{0}{0}\right)$ form

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 7x}{x} &= \lim_{x \rightarrow 0} \frac{\sin 7x}{7x} \times 7 \\ &= 1 \times 7 = 7 \quad \text{Ans.} \end{aligned}$$

(ii) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$ $\left(\frac{0}{0}\right)$ form

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} &= \lim_{x \rightarrow 0} \frac{\sin \frac{x\pi}{180}}{\frac{x\pi}{180}} \times \frac{\pi}{180} \\ &= 1 \times \frac{\pi}{180} = \frac{\pi}{180} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \therefore 1^\circ &= \frac{\pi}{180} \text{ radian} \\ x^\circ &= \frac{x\pi}{180} \text{ radian} \end{aligned}$$

(iii) $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta}$ $\left(\frac{0}{0}\right)$ form

$$\begin{aligned} \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta} &= \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta} \\ &= \lim_{\theta \rightarrow 0} \frac{1 - \cos^2 \theta}{\sin \theta (1 + \cos \theta)} \\ &= \lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\sin \theta (1 + \cos \theta)} \\ &= \lim_{\theta \rightarrow 0} \frac{\sin \theta}{1 + \cos \theta} \\ &= \frac{0}{1 + 1} = \frac{0}{2} = 0 \quad \text{Ans.} \end{aligned}$$



(iv) $\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x} \left(\frac{0}{0} \right) \text{form}$

Put $\pi - x = t \Rightarrow x = \pi - t$

As $x \rightarrow \pi$, $t \rightarrow 0$

$$\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x} = \lim_{t \rightarrow 0} \frac{\sin(\pi - t)}{t}$$

$$= \lim_{t \rightarrow 0} \frac{\sin t}{t}$$

$$= 1 \quad \text{Ans.}$$

(v) $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} \left(\frac{0}{0} \right) \text{form} \quad (G.B 2007)$

$$\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \lim_{x \rightarrow 0} \frac{\frac{\sin ax}{ax} \times ax}{\frac{\sin bx}{bx} \times bx}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin ax}{ax} \times a}{\frac{\sin bx}{bx} \times b}$$

$$= \frac{1 \times a}{1 \times b} = \frac{a}{b} \quad \text{Ans.}$$

(vi) $\lim_{x \rightarrow 0} \frac{x}{\tan x} \left(\frac{0}{0} \right) \text{form} \quad (L.B 2008)$

$$\lim_{x \rightarrow 0} \frac{x}{\tan x} = \lim_{x \rightarrow 0} \frac{x}{\frac{\sin x}{\cos x}}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x}{\frac{\sin x}{x}}$$

$$= \frac{1}{1} = 1 \quad \text{Ans.}$$

(vii) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} \left(\frac{0}{0} \right) \text{form}$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2} \quad (\because \cos 2x = 1 - 2\sin^2 x \Rightarrow 2\sin^2 x = 1 - \cos 2x)$$



$$\begin{aligned}
 &= \lim_{x \rightarrow 0} 2 \left(\frac{\sin x}{x} \right)^2 \\
 &= 2(1)^2 = 2 \quad \text{Ans.}
 \end{aligned}$$

(viii) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x} \left(\frac{0}{0} \right) \text{ form}$

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x} &= \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \cos^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{1 - \cos x}{(1 + \cos x)(1 - \cos x)} \\
 &= \lim_{x \rightarrow 0} \frac{1}{1 + \cos x} \\
 &= \frac{1}{1 + 1} = \frac{1}{2} \quad \text{Ans.}
 \end{aligned}$$

(ix) $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta} \left(\frac{0}{0} \right) \text{ form}$

$$\begin{aligned}
 \lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta} &= \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \times \sin \theta \\
 &= 1 \times 0 = 0 \quad \text{Ans.}
 \end{aligned}$$

(x) $\lim_{x \rightarrow 0} \frac{\sec x - \cos x}{x} \left(\frac{0}{0} \right) \text{ form} \quad (G.B 2007)$

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\sec x - \cos x}{x} &= \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} - \cos x}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\frac{1 - \cos^2 x}{\cos x}}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x \cos x} \\
 &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\sin x}{\cos x} \\
 &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \tan x = 1 \times 0 = 0 \quad \text{Ans.}
 \end{aligned}$$

(xi) $\lim_{\theta \rightarrow 0} \frac{1 - \cos p\theta}{1 - \cos q\theta} \left(\frac{0}{0} \right) \text{ form} \quad (G.B 2006)$

We know that:



$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos \theta = 1 - 2 \sin^2 \frac{\theta}{2}$$

$$\cos p\theta = 1 - 2 \sin^2 \frac{p\theta}{2}$$

$$2 \sin^2 \frac{p\theta}{2} = 1 - \cos p\theta \quad \text{and} \quad 2 \sin^2 \frac{q\theta}{2} = 1 - \cos q\theta$$

$$\begin{aligned} \lim_{\theta \rightarrow 0} \frac{1 - \cos p\theta}{1 - \cos q\theta} &= \lim_{\theta \rightarrow 0} \frac{2 \sin^2 \frac{p\theta}{2}}{2 \sin^2 \frac{q\theta}{2}} \\ &= \lim_{\theta \rightarrow 0} \frac{\left(\frac{\sin \frac{p\theta}{2}}{\frac{p\theta}{2}} \times \frac{p\theta}{2} \right)^2}{\left(\frac{\sin \frac{q\theta}{2}}{\frac{q\theta}{2}} \times \frac{q\theta}{2} \right)^2} \\ &= \lim_{\theta \rightarrow 0} \frac{\left(\frac{\sin \frac{p\theta}{2}}{\frac{p\theta}{2}} \right)^2 \cdot \frac{p^2 \theta^2}{4}}{\left(\frac{\sin \frac{q\theta}{2}}{\frac{q\theta}{2}} \right)^2 \cdot \frac{q^2 \theta^2}{4}} = \lim_{\theta \rightarrow 0} \frac{\left(\frac{\sin \frac{p\theta}{2}}{\frac{p\theta}{2}} \right)^2 \cdot p^2}{\left(\frac{\sin \frac{q\theta}{2}}{\frac{q\theta}{2}} \right)^2 \cdot q^2} \\ &= \frac{(1)^2 \cdot p^2}{(1)^2 \cdot q^2} = \frac{p^2}{q^2} \quad \text{Ans.} \end{aligned}$$

(xii) $\lim_{\theta \rightarrow 0} \frac{\tan \theta - \sin \theta}{\sin^3 \theta} \left(\frac{0}{0} \right) \text{ form } (L.B \ 2005)$

$$\lim_{\theta \rightarrow 0} \frac{\tan \theta - \sin \theta}{\sin^3 \theta} = \lim_{\theta \rightarrow 0} \frac{\frac{\sin \theta}{\cos \theta} - \sin \theta}{\sin^3 \theta}$$



$$\begin{aligned}
 &= \lim_{\theta \rightarrow 0} \frac{\sin\theta \left(\frac{1}{\cos\theta} - 1 \right)}{\sin^3\theta} \\
 &= \lim_{\theta \rightarrow 0} \frac{\frac{1 - \cos\theta}{\cos\theta}}{\sin^2\theta} \\
 &= \lim_{\theta \rightarrow 0} \frac{1 - \cos\theta}{\cos\theta (1 - \cos^2\theta)} \\
 &= \lim_{\theta \rightarrow 0} \frac{1 - \cos\theta}{\cos\theta(1 + \cos\theta)(1 - \cos\theta)} \\
 &= \lim_{\theta \rightarrow 0} \frac{1}{\cos\theta(1 + \cos\theta)} = \frac{1}{1(1 + 1)} = \frac{1}{2} \quad \text{Ans.}
 \end{aligned}$$

Q.4 Express each limit in terms of e:

- (i) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^{2n}$ (ii) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^{n/2}$
 (iii) $\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n}\right)^n$ (iv) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{3n}\right)^n$
 (v) $\lim_{n \rightarrow +\infty} \left(1 + \frac{4}{n}\right)^n$ (vi) $\lim_{x \rightarrow 0} (1 + 3x)^{2/x}$
 (vii) $\lim_{x \rightarrow 0} (1 + 2x^2)^{1/x^2}$ (viii) $\lim_{h \rightarrow 0} (1 - 2h)^{1/h}$
 (ix) $\lim_{x \rightarrow \infty} \left(\frac{x}{1+x}\right)^x$ (L.B 2003,04) (x) $\lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}, \quad x < 0$
 (xi) $\lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}, \quad x > 0$ (L.B 2005)

Solution:

- (i) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^{2n} = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{n}\right)^n\right]^2$
 $= e^2$ Ans.
 (ii) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^{n/2} = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{n}\right)^n\right]^{1/2}$
 $= e^{1/2}$ Ans.



$$(iii) \quad \lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n}\right)^n = \lim_{n \rightarrow +\infty} \left[\left(1 + \left(\frac{1}{-n}\right)\right)^{-n}\right]^{-1} \\ = e^{-1} \quad \text{Ans.}$$

$$(iv) \quad \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{3n}\right)^n = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{3n}\right)^{3n}\right]^{1/3} \\ = e^{1/3} \quad \text{Ans.}$$

$$(v) \quad \lim_{n \rightarrow +\infty} \left(1 + \frac{4}{n}\right)^n = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{n/4}\right)^{n/4}\right]^4 \\ = e^4 \quad \text{Ans.}$$

$$(vi) \quad \lim_{n \rightarrow 0} (1 + 3x)^{2/x} = \lim_{n \rightarrow 0} [(1 + 3x)^{1/3x}]^{2 \times 3} \\ = e^6 \quad \text{Ans.}$$

$$(vii) \quad \lim_{x \rightarrow 0} (1 + 2x^2)^{1/x^2} = \lim_{x \rightarrow 0} [(1 + 2x^2)^{1/2x^2}]^2 \\ = e^2 \quad \text{Ans.}$$

$$(viii) \quad \lim_{h \rightarrow 0} (1 - 2h)^{1/h} = \lim_{h \rightarrow 0} [(1 + (-2h)^{-1/2h})^{-2}] \\ = e^{-2} \quad \text{Ans.}$$

$$(ix) \quad \lim_{x \rightarrow \infty} \left(\frac{x}{1+x}\right)^x = \lim_{x \rightarrow \infty} \left(\frac{1+x}{x}\right)^{-x} \quad (G.B 2006) (L.B 2007) \\ = \lim_{x \rightarrow \infty} \left(\frac{1}{x} + \frac{x}{x}\right)^{-x} \\ = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{x}\right)^x\right]^{-1} = e^{-1} \quad \text{Ans.}$$

$$(x) \quad \lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}, \quad x < 0 \quad (G.B 2005)$$

Put, $x = -t$, where $t > 0$

As, $x \rightarrow 0$, $t \rightarrow 0$

$$\lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1} = \lim_{t \rightarrow 0} \frac{e^{1/-t} - 1}{e^{1/-t} + 1} \\ = \frac{e^{-1/0} - 1}{e^{-1/0} + 1} = \frac{e^{-\infty} - 1}{e^{-\infty} + 1} \\ = \frac{0 - 1}{0 + 1} = \frac{-1}{1} = -1 \quad \text{Ans.}$$



$$(xi) \quad \lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}, \quad x > 0 \quad \left(\frac{\infty}{\infty} \right)$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1} &= \lim_{x \rightarrow 0} \frac{e^{1/x} \left(1 - \frac{1}{e^{1/x}} \right)}{e^{1/x} \left(1 + \frac{1}{e^{1/x}} \right)} \\ &= \lim_{x \rightarrow 0} \frac{1 - \frac{1}{e^{1/x}}}{1 + \frac{1}{e^{1/x}}} \\ &= \frac{1 - \frac{1}{e^{\infty}}}{1 + \frac{1}{e^{\infty}}} \\ &= \frac{1 - \frac{1}{\infty}}{1 + \frac{1}{\infty}} = \frac{1 - 0}{1 + 0} = 1 \quad \text{Ans.} \end{aligned}$$

Continuous Function

A function f is said to be continuous at a number “c” if and only if the following three conditions are satisfied.

- (i) $f(c)$ is defined.
- (ii) $\lim_{x \rightarrow c} f(x)$ exists.
- (iii) $\lim_{x \rightarrow c} f(x) = f(c)$



EXERCISE 1.4

Q.1 Determine the left hand limit and right hand limit and then find limits of the following functions at $x = c$.

(i) $f(x) = 2x^2 + x - 5$, $c = 1$

(ii) $f(x) = \frac{x^2 - 9}{x - 3}$, $c = -3$

(iii) $f(x) = |x - 5|$, $c = 5$

Solution:

(i) $f(x) = 2x^2 + x - 5$, $c = 1$
Left hand limit



$$\begin{aligned}\lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (2x^2 + x - 5) \\ &= 2(1)^2 + 1 - 5 \\ &= 2 - 4 = -2 \quad \text{Ans.}\end{aligned}$$

Right hand limit

$$\begin{aligned}\lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (2x^2 + x - 5) \\ &= 2(1)^2 + 1 - 5 \\ &= 2 + 1 - 5 \\ &= -2 \quad \text{Ans.}\end{aligned}$$

(ii) $f(x) = \frac{x^2 - 9}{x - 3}$, $c = -3$

Left hand limit

$$\begin{aligned}\lim_{x \rightarrow -3^-} f(x) &= \lim_{x \rightarrow -3^-} \frac{x^2 - 9}{x - 3} \\ &= \lim_{x \rightarrow -3^-} \frac{(x + 3)(x - 3)}{x - 3} \\ &= \lim_{x \rightarrow -3^-} (x + 3) \\ &= -3 + 3 = 0 \quad \text{Ans.}\end{aligned}$$

Right hand limit

$$\begin{aligned}\lim_{x \rightarrow -3^+} f(x) &= \lim_{x \rightarrow -3^+} \frac{x^2 - 9}{x - 3} \\ &= \lim_{x \rightarrow -3^+} \frac{(x + 3)(x - 3)}{x - 3} \\ &= \lim_{x \rightarrow -3^+} (x + 3) \\ &= -3 + 3 = 0 \quad \text{Ans.}\end{aligned}$$

(iii) $f(x) = |x - 5|$, $c = 5$

Left hand limit

$$\begin{aligned}\lim_{x \rightarrow 5^-} f(x) &= \lim_{x \rightarrow 5^-} |x - 5| \\ &= \lim_{x \rightarrow 5^-} -(x - 5) \\ &= -(5 - 5) = 0 \quad \text{Ans.}\end{aligned}$$



Right hand limit

$$\begin{aligned}\lim_{x \rightarrow 5^+} f(x) &= \lim_{x \rightarrow 5^+} |x - 5| \\ &= \lim_{x \rightarrow 5^+} (x - 5) \\ &= 5 - 5 \\ &= 0 \quad \text{Ans.}\end{aligned}$$

Q.2 Discuss the continuity of $f(x)$ at $x = c$:

(i) $f(x) = \quad$, $c = 2$

(G.B 2007, L.B 2008)

(ii) $f(x) = \begin{cases} 3x - 1 & \text{if } x < 1 \\ 4 & \text{if } x = 1 \\ 2x & \text{if } x > 1 \end{cases}$, $c = 1$

(L.B 2009, L.B 2006)

(L.B 2009, G.B 2007)

Solution:

(i) $f(x) = \begin{cases} 2x + 5 & \text{if } x \leq 2 \\ 4x + 1 & \text{if } x > 2 \end{cases}$, $c = 2$

$$\begin{aligned}f(2) &= 2(2) + 5 \\ &= 4 + 5 \\ &= 9\end{aligned}$$

Left hand limit

$$\begin{aligned}\lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} (2x + 5) \\ &= 2(2) + 5 \\ &= 4 + 5 = 9\end{aligned}$$

Right hand limit

$$\begin{aligned}\lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^+} (4x + 1) \\ &= 4(2) + 1 \\ &= 8 + 1 \\ &= 9\end{aligned}$$

$\therefore$ Left hand limit = Right hand limit

So $\lim_{x \rightarrow 2} f(x)$ exists

$$\therefore f(2) = \lim_{x \rightarrow 2} f(x) = 9$$

So the function is continuous at $x = 2$.



$$(ii) \quad f(x) = \begin{cases} 3x - 1 & \text{if } x < 1 \\ 4 & \text{if } x = 1 \\ 2x & \text{if } x > 1 \end{cases}, \quad c = 1 \quad (L.B \text{ 2006, 2007})$$

$$f(1) = 4$$

Left hand limit

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (3x - 1) \\ &= 3(1) - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

Right hand limit

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (2x) \\ &= 2(1) = 2 \end{aligned}$$

$\therefore$ Left hand limit = Right hand limit

So $\lim_{x \rightarrow 1} f(x)$ exists

$$\therefore f(1) \neq \lim_{x \rightarrow 1} f(x)$$

So the function is discontinuous at $x = 1$.

$$Q.3 \quad \text{If } f(x) = \begin{cases} 3x & \text{if } x \leq -2 \\ x^2 - 1 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \geq 2 \end{cases} \quad (L.B \text{ 2011})$$

Discuss continuity at $x = 2$ and $x = -2$.

Solution:

$$\text{At } x = 2$$

$$f(2) = 3$$

Left hand limit

$$\begin{aligned} \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} (x^2 - 1) \\ &= 2^2 - 1 = 4 - 1 = 3 \end{aligned}$$

Right hand limit

$$\begin{aligned} \lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^+} 3 \\ &= 3 \end{aligned}$$

$\therefore$ Left hand limit = Right hand limit

So, $\lim_{x \rightarrow 2} f(x)$ exists



$$\therefore f(2) = \lim_{x \rightarrow 2} f(x) = 3$$

So the function is continuous at $x = 2$.

At $x = -2$

$$f(-2) = 3(-2) = -6$$

Left hand limit.

$$\begin{aligned} \lim_{x \rightarrow -2^-} f(x) &= \lim_{x \rightarrow -2^-} (3x) \\ &= 3(-2) = -6 \end{aligned}$$

Right hand limit.

$$\begin{aligned} \lim_{x \rightarrow -2^+} f(x) &= \lim_{x \rightarrow -2^+} (x^2 - 1) \\ &= (-2)^2 - 1 \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

$\therefore$ Left hand limit $\neq$ Right hand limit

So, $\lim_{x \rightarrow -2} f(x)$ does not exist.

$$\therefore f(-2) \neq \lim_{x \rightarrow -2} f(x)$$

So the function is discontinuous at $x = -2$.

Q.4 If $f(x) = \begin{cases} x+2, & x \leq -1 \\ c+2, & x > -1 \end{cases}$ find 'c' so that $\lim_{x \rightarrow -1} f(x)$ exists. (L.B 2009 Supply)
(G.B 2008)

Solution:

Left hand limit

$$\begin{aligned} \lim_{x \rightarrow -1^-} f(x) &= \lim_{x \rightarrow -1^-} (x+2) \\ &= -1+2 = 1 \end{aligned}$$

Right hand limit

$$\begin{aligned} \lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} (c+2) \\ &= c+2 \end{aligned}$$

Since $\lim_{x \rightarrow -1} f(x)$ exists.

$\therefore$ Left hand limit = Right hand limit

$$1 = c+2$$

$$c = 1-2$$

$$\boxed{c = -1}$$

Ans.



Q.5 Find the values m and n, So that given function f is continuous at x = 3:

$$(i) \quad f(x) = \begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x + 9 & \text{if } x > 3 \end{cases} \quad (ii) \quad f(x) = \begin{cases} mx & \text{if } x < 3 \\ x^2 & \text{if } x \geq 3 \end{cases}$$

Solution:

$$(i) \quad f(x) = \begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x + 9 & \text{if } x > 3 \end{cases} \quad \begin{matrix} (L.B \ 2004, \ 2005) \\ (G.B \ 2006, \ 2009) \end{matrix}$$

$$f(3) = n$$

Left hand limit

$$\begin{aligned} \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} (mx) \\ &= 3m \end{aligned}$$

Right hand limit

$$\begin{aligned} \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} (-2x + 9) \\ &= -2(3) + 9 \\ &= -6 + 9 \\ &= 3 \end{aligned}$$

Since f(x) is continuous at x = 3

$$\therefore \text{Left hand limit} = \text{Right hand limit} = f(3)$$

$$\begin{aligned} 3m &= 3 = n \\ 3m &= 3, \quad 3 = n \\ m &= \frac{3}{3} \quad n = 3 \\ m &= 1 \end{aligned}$$

$$\therefore m = 1, \quad n = 3 \quad \text{Ans.}$$

$$(ii) \quad f(x) = \begin{cases} mx & \text{if } x < 3 \\ x^2 & \text{if } x \geq 3 \end{cases} \quad (L.B \ 2007)$$

$$f(3) = (3)^2 = 9$$

Left hand limit

$$\begin{aligned} \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} (mx) \\ &= 3m \end{aligned}$$

Right hand limit

$$\begin{aligned} \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} (x^2) \\ &= 3^2 = 9 \end{aligned}$$

Since f(x) is continuous at x = 3



$$\begin{aligned}
 \therefore \quad \text{Left hand limit} &= \text{Right hand limit} = f(3) \\
 3m &= 9 = 9 \\
 3m &= 9 \\
 m &= \frac{9}{3} = 3 \quad \text{Ans.}
 \end{aligned}$$

Q.6: If $f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2}, & x \neq 2 \\ k, & x = 2 \end{cases}$

(G.B 2004)
(L.B 2009 (s) 2004)
(G.B 2006)
(L.B 2008)
(G.B 2008)

Find value of k so that f is continuous at $x = 2$.

Solution:

$$f(2) = k$$

$$\begin{aligned}
 \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} \left(\frac{0}{0} \right) \text{ form} \\
 &= \lim_{x \rightarrow 2} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} \times \frac{\sqrt{2x+5} + \sqrt{x+7}}{\sqrt{2x+5} + \sqrt{x+7}} \\
 &= \lim_{x \rightarrow 2} \frac{(\sqrt{2x+5})^2 - (\sqrt{x+7})^2}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})} \\
 &= \lim_{x \rightarrow 2} \frac{(2x+5) - (x+7)}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})} \\
 &= \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})} \\
 &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{2x+5} + \sqrt{x+7}} \\
 &= \frac{1}{\sqrt{2(2)+5} + \sqrt{2+7}} \\
 &= \frac{1}{\sqrt{4+5} + \sqrt{9}} \\
 &= \frac{1}{3+3} = \frac{1}{6}
 \end{aligned}$$

Since $f(x)$ is continuous at $x = 2$

$$\therefore f(2) = \lim_{x \rightarrow 2} f(x)$$

$$\boxed{k = \frac{1}{6}} \quad \text{Ans.}$$



EXERCISE 1.5

Q.1 Draw the graphs of the following equations.

(i) $x^2 + y^2 = 9$

(ii) $\frac{x^2}{16} + \frac{y^2}{4} = 1$

(iii) $y = e^{2x}$

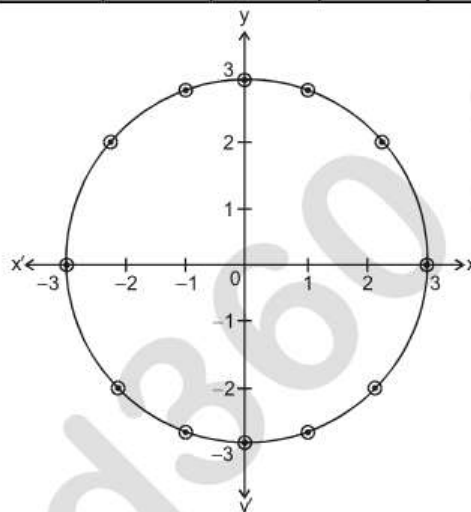
(iv) $y = 3^x$

Solution:

(i) $x^2 + y^2 = 9$
 $y^2 = 9 - x^2$
 $y = \pm \sqrt{9 - x^2}$

Its domain is $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
$y = \pm \sqrt{9 - x^2}$	0	± 2.2	± 2.8	± 3	± 2.8	± 2.2	0

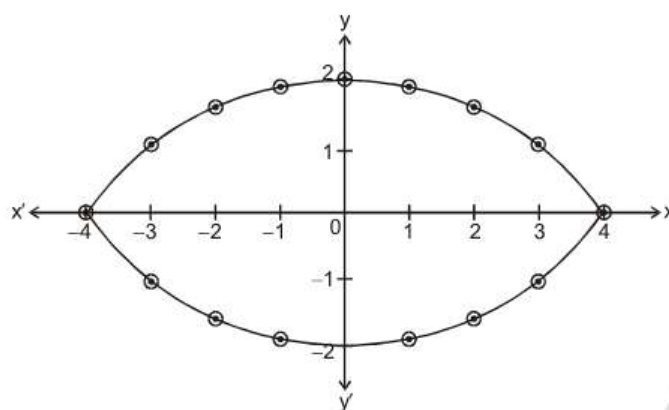


(ii) $\frac{x^2}{16} + \frac{y^2}{4} = 1$
 $\frac{y^2}{4} = 1 - \frac{x^2}{16}$
 $y^2 = 4 \left(\frac{16 - x^2}{16} \right)$
 $y^2 = \frac{16 - x^2}{4}$
 $y = \pm \frac{\sqrt{16 - x^2}}{2}$

Its domain is $-4 \leq x \leq 4$.

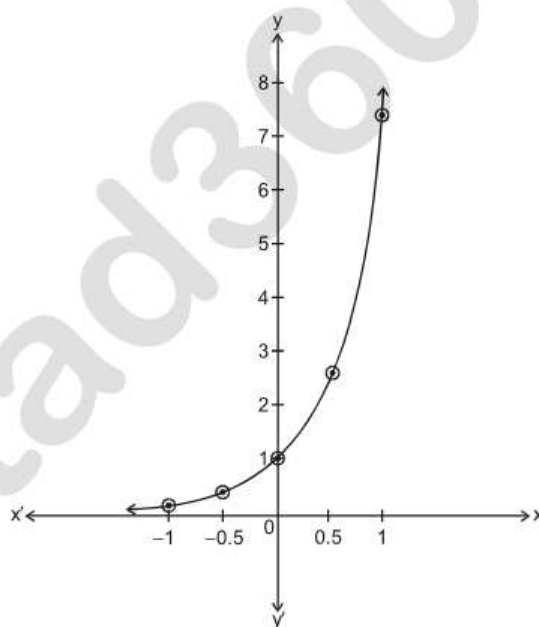


x	-4	-3	-2	-1	0	1	2	3	4
$y = \pm \frac{\sqrt{9-x^2}}{2}$	0	± 1.3	± 1.7	± 1.9	± 2	± 1.9	± 1.7	± 1.3	0



(iii) $y = e^{2x}$

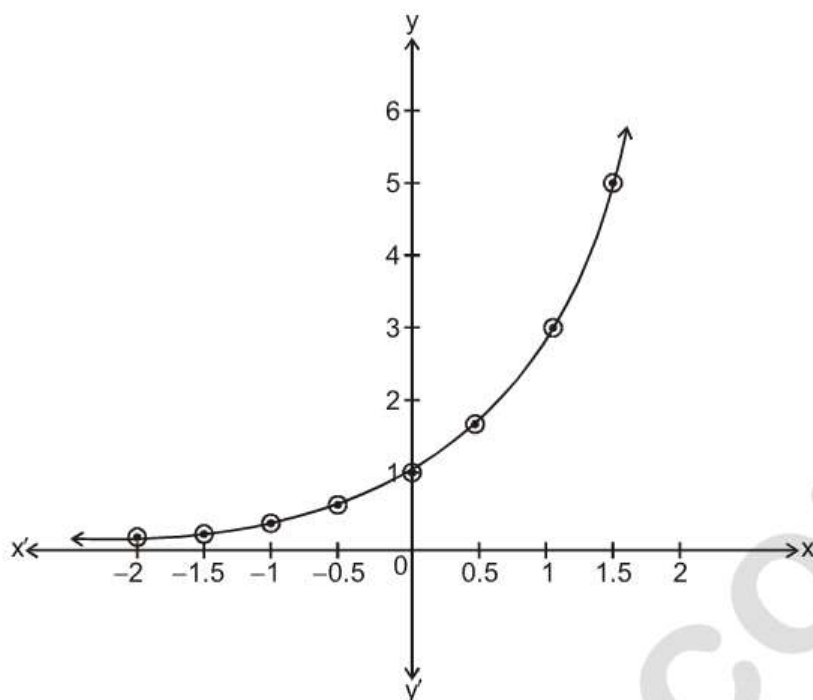
x	-1	-0.5	0	0.5	1
$y = e^{2x}$	0.1	0.4	1	2.7	7.4



(iv) $y = 3^x$

x	-2	-1.5	-1	-0.5	0	0.5	1	1.5
$y = 3^x$	0.1	0.2	0.3	0.6	1	1.7	3	5.2





Q.2 Graph the curves that has the parametric equations given below.

(i) $x = t$, $y = t^2$, $-3 \leq t \leq 3$ where 't' is a parameter

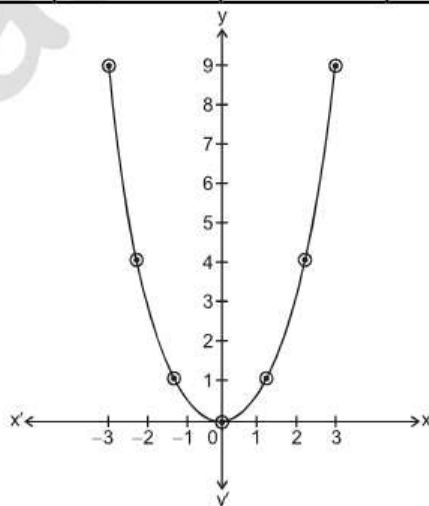
(ii) $x = t - 1$, $y = 2t - 1$, $-1 < t < 5$ where 't' is a parameter

(iii) $x = \sec\theta$, $y = \tan\theta$ where ' θ ' is a parameter

Solution:

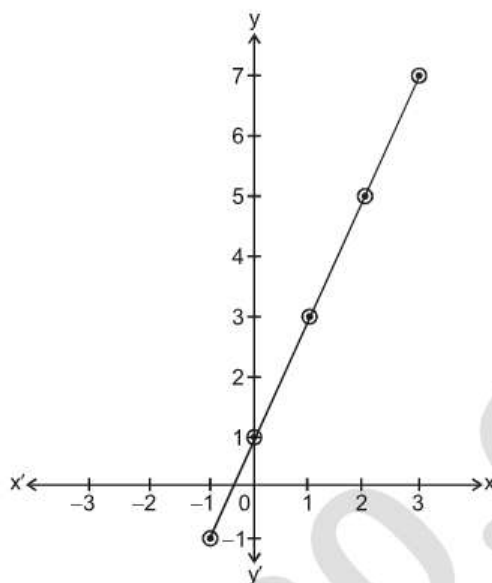
(i) $x = t$, $y = t^2$, $-3 \leq t \leq 3$ where 't' is a parameter

t	-3	-2	-1	0	1	2	3
$x = t$	-3	-2	-1	0	1	2	3
$y = t^2$	9	4	1	0	1	4	9



(ii) $x = t - 1$, $y = 2t - 1$, $-1 < t < 5$ where 't' is a parameter

t	0	1	2	3	4
$x = t - 1$	-1	0	1	2	3
$y = 2t - 1$	-1	1	3	5	7



(iii) $x = \sec\theta$, $y = \tan\theta$ where 'θ' is a parameter

$$x^2 = \sec^2\theta, \quad y^2 = \tan^2\theta$$

$$x^2 - y^2 = \sec^2\theta - \tan^2\theta$$

$$x^2 - y^2 = 1$$

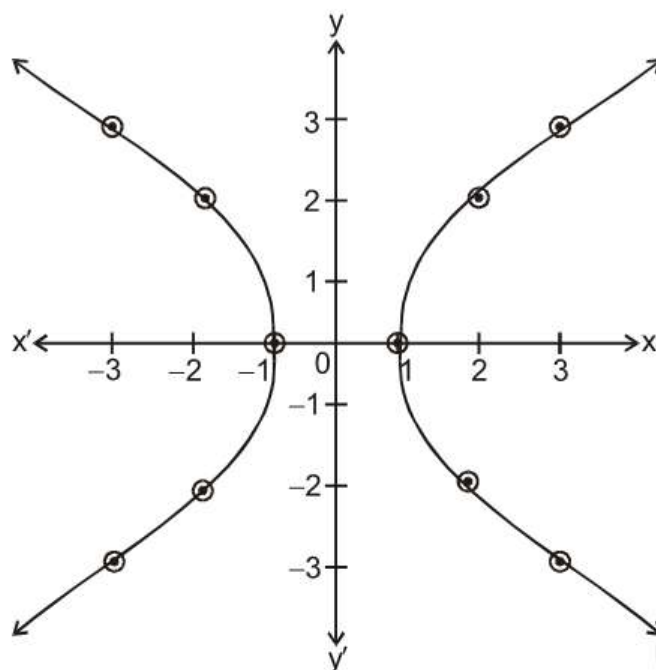
$$(\because 1 + \tan^2\theta = \sec^2\theta \Rightarrow 1 = \sec^2\theta - \tan^2\theta)$$

$$y^2 = x^2 - 1$$

$$y = \pm \sqrt{x^2 - 1}$$

x	-3	-2	-1	1	2	3
$y = \sqrt{x^2 - 1}$	± 2.8	± 1.7	0	0	± 1.7	± 2.8





Q.3 Draw the graphs of the functions defined below and find whether they are continuous.

(i) $y = \begin{cases} x-1 & \text{if } x < 3 \\ 2x+1 & \text{if } x \geq 3 \end{cases}$

(ii) $y = \frac{x^2-4}{x-2}, \quad x \neq 2$

(iii) $y = \begin{cases} x+3 & , \quad x \neq 3 \\ 2 & , \quad x = 3 \end{cases}$

(iv) $y = \frac{x^2-16}{x-4}, \quad x \neq 4$

Solution:

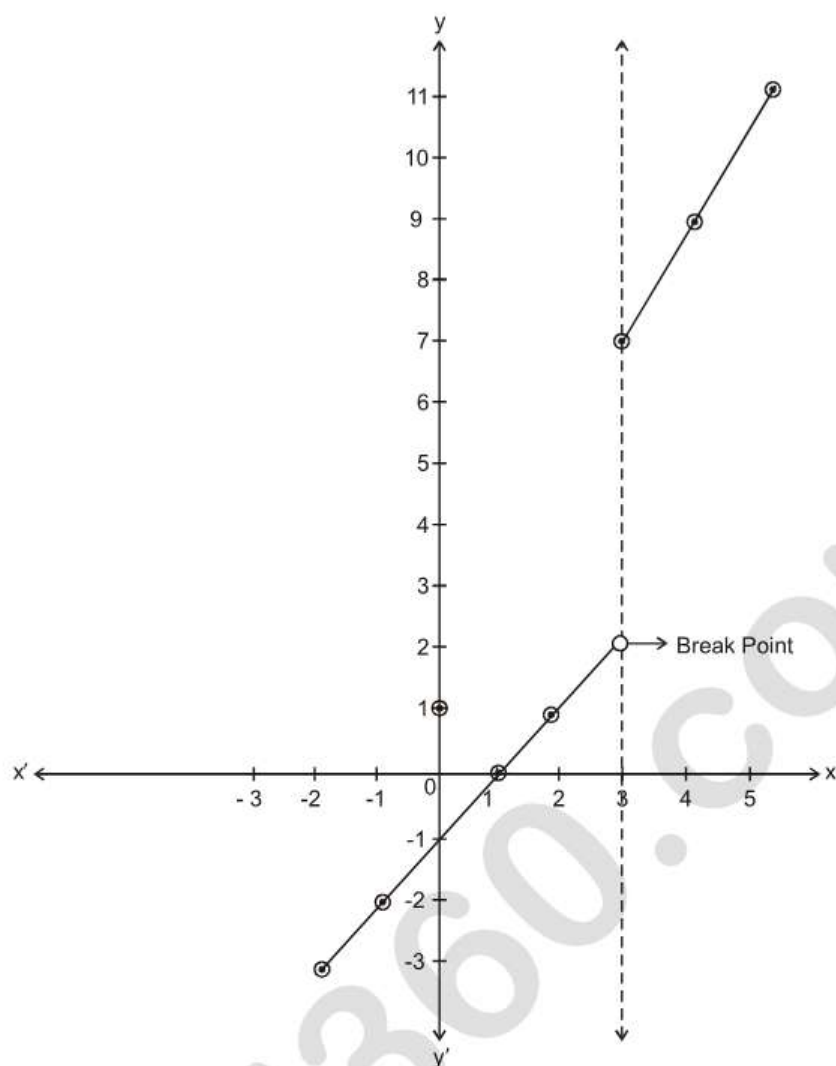
(i) $y = \begin{cases} x-1 & \text{if } x < 3 \\ 2x+1 & \text{if } x \geq 3 \end{cases}$
 $y = x-1, \quad x < 3$

x	-2	-1	0	1	2
$y = x - 1$	-3	-2	-1	0	1

$y = 2x + 1, \quad x \geq 3$

x	3	4	5
$y = 2x + 1$	7	9	11



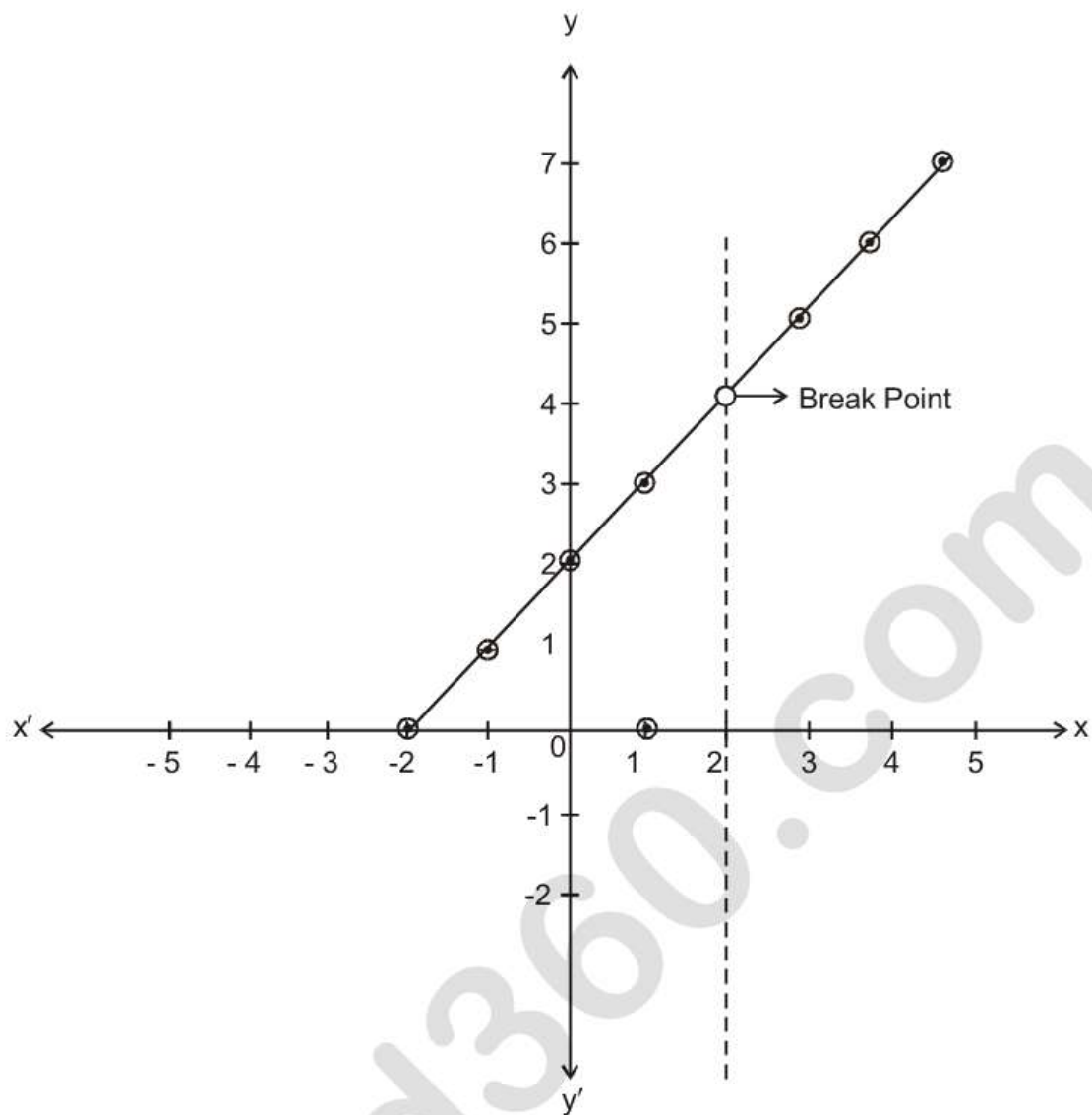


Since there is a break in a graph. So this function is not continuous.

$$\begin{aligned}
 \text{(ii)} \quad y &= \frac{x^2 - 4}{x - 2}, \quad x \neq 2 \\
 &= \frac{(x + 2)(x - 2)}{x - 2}, \quad x \neq 2 \\
 y &= x + 2, \quad x \neq 2
 \end{aligned}$$

x	-3	-2	-1	0	1	3	4	5
y	-1	0	1	2	3	5	6	7





Since there is a break in a graph so this function is not continuous.

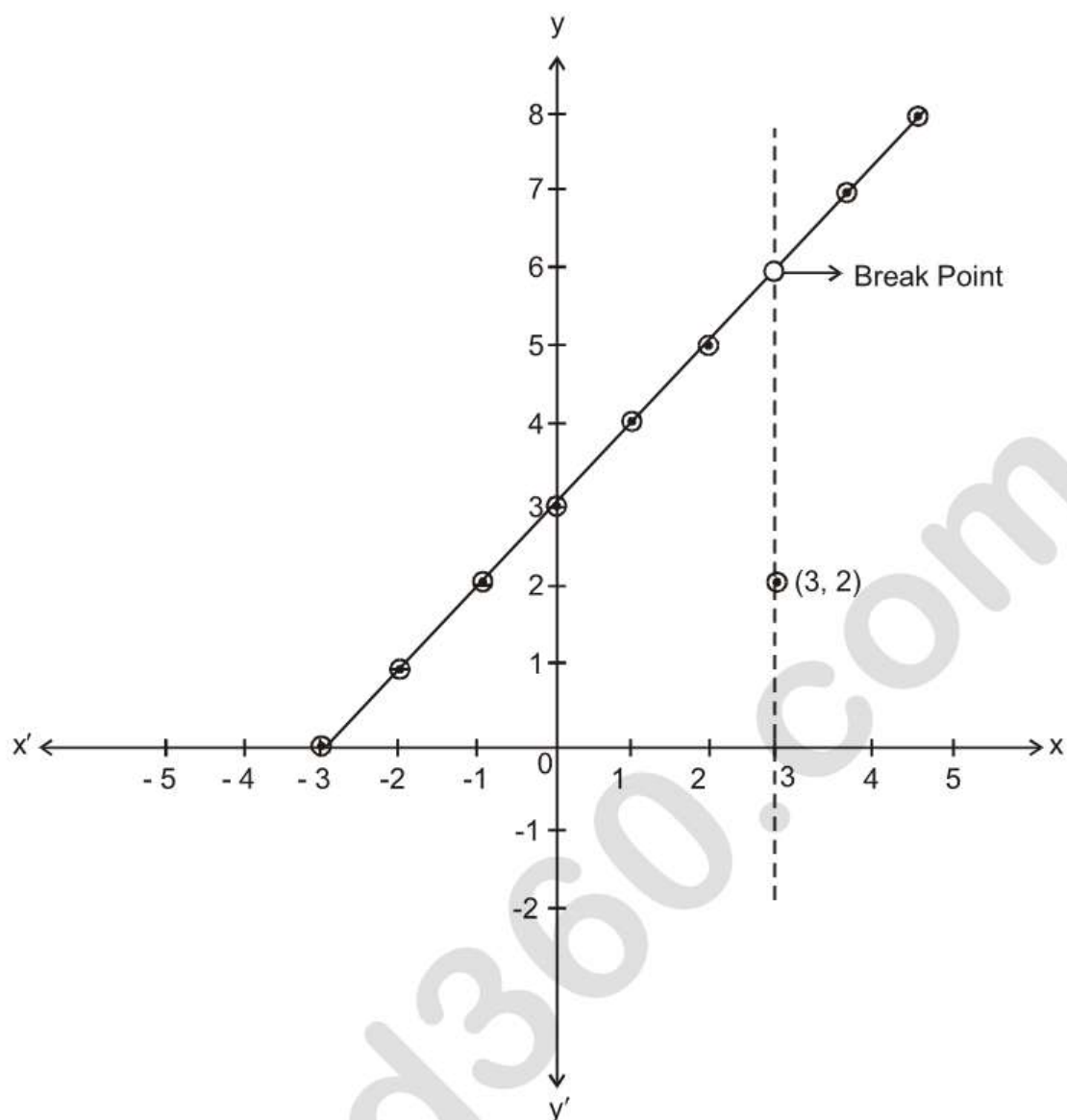
$$(iii) \quad y = \begin{cases} x + 3 & \text{if } x \neq 3 \\ 2 & \text{if } x = 3 \end{cases}$$

$$y = x + 3 \quad \text{if } x \neq 3$$

x	-3	-2	-1	0	1	3	4	5
y	0	1	2	3	4	5	7	8

$$y = 2 \quad \text{if } x = 3$$

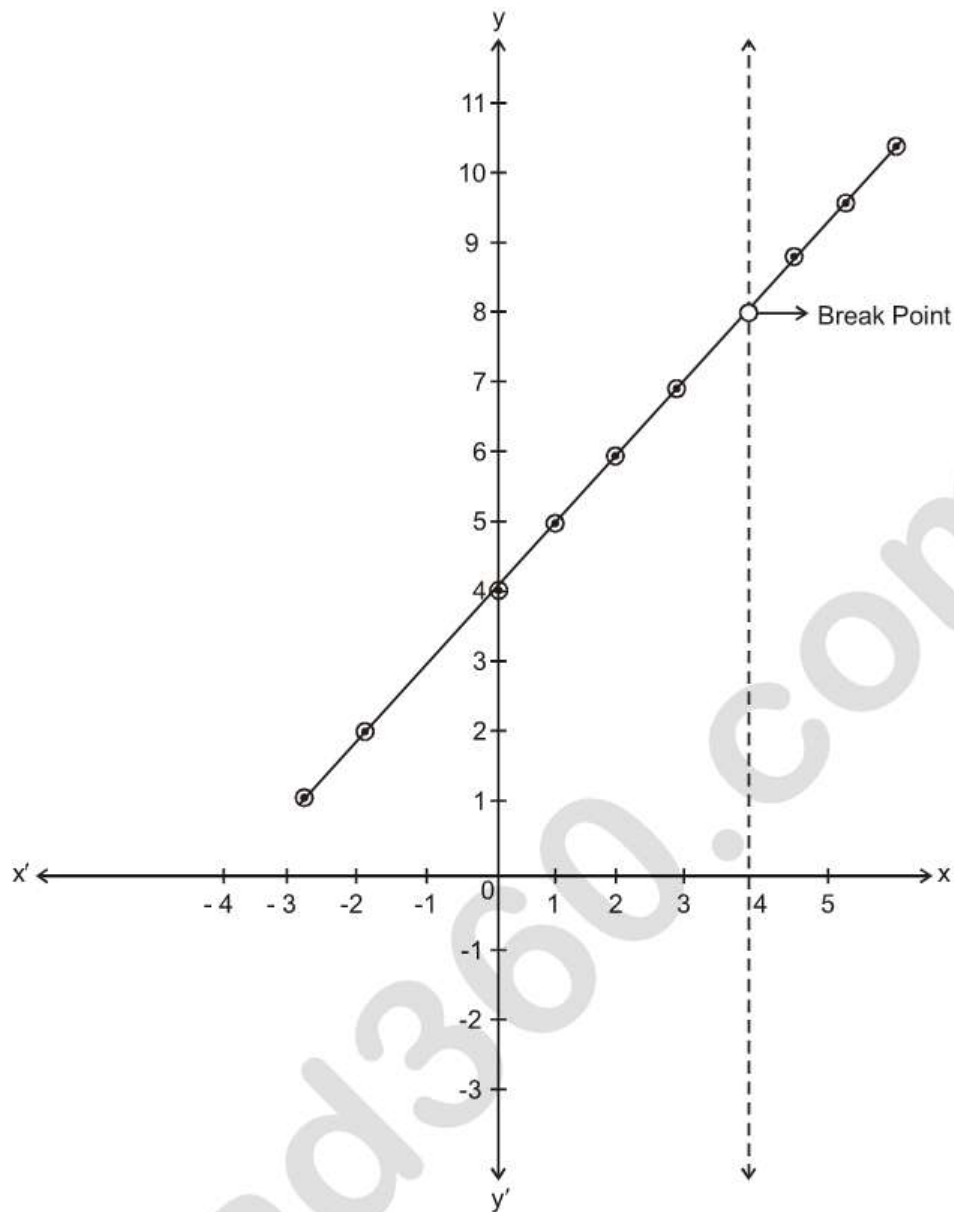




Since there is a break in a graph. So this function is not continuous at $x = 3$.

$$\begin{aligned}
 \text{(iv)} \quad y &= \frac{x^2 - 16}{x - 4}, \quad x \neq 4 \\
 &= \frac{(x + 4)(x - 4)}{x - 4}, \quad x \neq 4
 \end{aligned}$$

x	-3	-2	-1	0	1	2	3	5	6
y	1	2	3	4	5	6	7	9	10



Since there is a break in a graph. So this function is not continuous at $x = 4$.

Q.4 Find the graphical solution of the following equations.

(i) $x = \sin 2x$

(ii) $\frac{x}{2} = \cos x$

(iii) $2x = \tan x$

Solution:

(i) Let $y = x = \sin 2x$

Therefore $y = x$ and $y = \sin 2x$

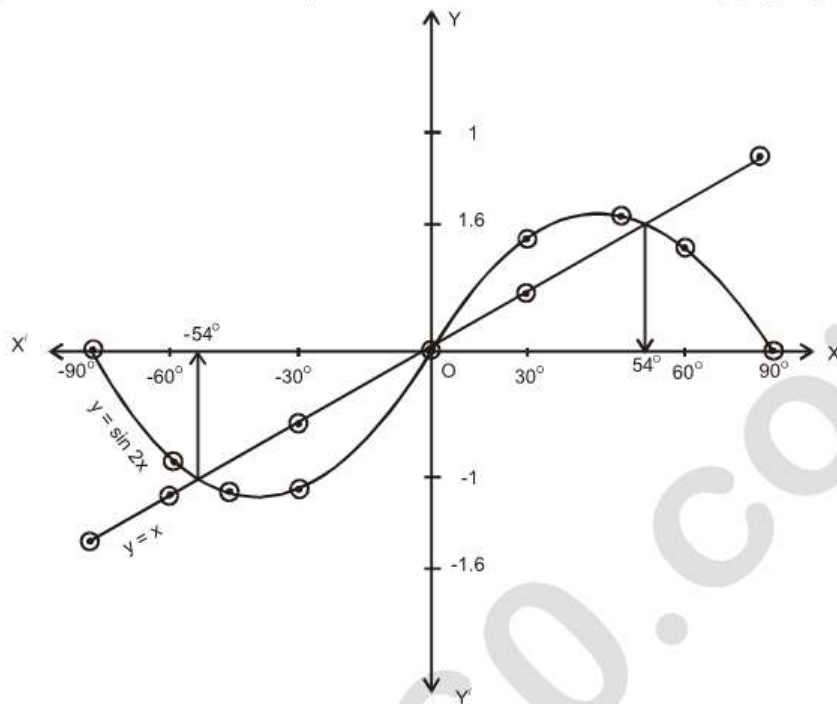
x	-90°	-60°	-30°	0°	30°	60°	90°
$y = x$	$-\pi/2 = -1.6$	$-\pi/3 = -1.05$	$-\pi/6 = -0.52$	0	$\pi/6 = 0.52$	$\pi/3 = 1.05$	$\pi/2 = 1.6$



$$y = \sin 2x$$

x	-90°	-60°	-30°	0°	30°	60°	90°
$y = \sin 2x$	0	-0.87	-0.87	0	0.87	0.87	0

The graphical solution is the points of intersection of two graphs, i.e. $x = 0^\circ, 54^\circ$



(ii) Let $y = \frac{x}{2} = \cos x$

Therefore $y = \frac{x}{2}$ and $y = \cos x$

$$y = \frac{x}{2}$$

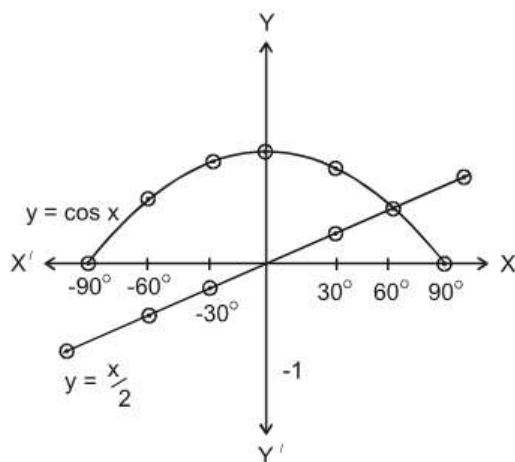
x	-90°	-60°	-30°	0°	30°	60°	90°
$y = \frac{x}{2}$	$-\pi/4$ $= -0.79$	$-\pi/6$ $= -0.52$	$-\pi/12$ $= -0.26$	0	$\pi/6$ $= 0.26$	$\pi/6$ $= 0.52$	$\pi/4$ $= 0.79$

$$y = \cos x$$

x	-90°	-60°	-30°	0°	30°	60°	90°
$y = \cos x$	0	0.5	0.87	1	0.87	0.5	0

The graphical solution is the point on x-axis, which is just below the point of intersection of two graphs. Hence $x = 60^\circ$.





(iii) Let $y = 2x = \tan x$
 Therefore $y = 2x$ and $y = \tan x$
 $y = 2x$

x	-90°	-60°	-30°	0°	30°	60°	90°
$y = 2x$	$-\pi = -3.14$	$-2\pi/3 = -2.09$	$-\pi/3 = -1.05$	0	$\pi/3 = 1.05$	$2\pi/3 = 2.09$	$\pi = 3.14$

$y = \tan x$

x	-90°	-60°	-30°	0°	30°	60°	90°
$y = \tan x$	∞	-1.73	-0.58	0	0.58	1.73	∞

The graphical solution is the point of intersection of two graphs, i.e. $x = 0^\circ$.

